On Variance Reduction in Learning Mean Flows

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Abstract

Mean Flows promise distillation-free, one-step generation by optimizing an identity between the average and instantaneous velocity field. In practice they suffer from a non-decreasing, high-variance loss whose root cause has been diagnosed by concurrent work as gradient conflict, Jacobian amplification, and a curvature bottleneck. We show that all three diagnoses are manifestations of a single quantity—the material derivative of the marginal velocity field—and call this the *curvature-variance principle*. From this principle we derive *Variance-Aware MeanFlow* (VaMF), a backbone-preserving training framework that combines an EMA velocity anchor (deterministic JVP tangent), a flow-matching anchor (boundary supervision), and a Hutchinson-estimated trace weight that downweights each sample by the local Jacobian-deviation norm $\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)$—a closed-form scalar approximation to the BLUE per-sample weight under isotropic conditional-velocity noise.

Add experiment results later...

1 Introduction

Deep generative models that leverage deep neural networks to learn and generate samples from unknown data distributions have achieved profound success. One of the predominant paradigms adopts flow-based generative models [1, 2, 3, 4, 5, 6, 7] that learn continuous-time transformations between a simple prior and a target data distribution. The existing conditional flow matching [5, 6] and stochastic interpolant [7, 8] enable simulation-free training by regressing a velocity field, achieving sample quality competitive with diffusion models [9, 10, 11, 12, 13, 14, 15] and variational autoencoders [16]. However, flow-based models share the same limitation as diffusion models as they require iterative integration at inference time, which can suffer from extensive computational costs and numerical instability.

The issue motivates consecutive studies on reducing the number of function evaluations to one or few steps [17]. Existing approaches include progressive distillation [18], distribution matching [19, 20], consistency models [21, 22], and flow map matching [23], but most require a pre-trained teacher model. Recently, Mean Flows [24] offers an appealing *distillation-free* alternative by learning a two-parameter average velocity field $\boldsymbol{u}(\boldsymbol{x}, r, t)$ through a total-derivative identity between the average and marginal velocity field. In principle, this identity provides exact self-supervision. The compound prediction $\boldsymbol{u} + (t-r)\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{u}$ should match the instantaneous velocity at convergence, requiring no external teacher. Nevertheless, empirical training of Mean Flows is plagued by non-decreasing loss and prohibitively high gradient variance [25, 26].

In this work, we conduct a theoretical analysis to showcase that these issues are *three manifestations of a single quantity*: the material derivative $\frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t} := \partial_t \boldsymbol{v} + (\nabla_{\mathbf{x}}\boldsymbol{v})\boldsymbol{v}$ of the marginal velocity field, which measures how rapidly the velocity changes along its own trajectories. We call this the **curvature-variance principle**. Specifically, we prove that: **(i)** the curvature gap $\Delta := \boldsymbol{u} - \boldsymbol{v}$ satisfies $\Delta \approx -\frac{(t-r)}{2}\frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t}$, so it is the local flow acceleration that causes the average and instantaneous velocities to diverge; **(ii)** the per-step gradient variance is amplified by a Jacobi factor $\mathrm{Tr}(\boldsymbol{J}\Sigma_{\boldsymbol{v}'}\boldsymbol{J}^\top)$ where $\boldsymbol{J} = (t-r)\partial_{\boldsymbol{z}}\boldsymbol{u}_{\boldsymbol{\theta}} - \boldsymbol{I}_d$, and the stop-gradient operator creates a semi-gradient gap that blinds the optimizer to this amplification; and **(iii)** the acceleration-to-noise ratio $\mathrm{ANR} = (t-r)\|\frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t}\|/\sqrt{\mathrm{Tr}(\Sigma_{\boldsymbol{v}'})}$ governs the optimal sampling distribution over intervals, connecting scheduling heuristics in prior work to a principled criterion.

From this analysis we derive **Variance-Aware MeanFlow (VaMF)**, a training framework that addresses all identified failure modes without architectural modification. First, we use an *EMA velocity anchor* in the total derivative to establish a deterministic JVP tangent, provably eliminating Jacobian variance amplification. We then introduce a *flow matching anchor*, an additional objective that supervises the boundary condition and prevents tangent bias. Finally, we apply a *trace weight* that downweights each sample by the local Jacobian-deviation norm $\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)$, which is a closed-form scalar approximation, estimated with a single Hutchinson probe, to the Best Linear Unbiased Estimator (BLUE) per-sample weight under the isotropic-noise approximation $\Sigma_{\boldsymbol{v}'} \approx \sigma_t^2 \boldsymbol{I}_d$. The trace weight subsumes inverse-variance weighting, progressive scheduling, and acceleration-aware interval sampling in a single per-sample scalar, while preserving the original MSE objective in Mean Flows. In summary, our contributions in this paper are as follows:

C1 We establish a theoretical principle that unifies gradient conflict, Jacobian amplification, and the curvature bottleneck under the curvature-variance principle, including the first formal quantification of the semi-gradient gap in learning Mean Flows.

C2 We derive the VaMF training framework that eliminates variance amplification and tangent bias through introducing EMA anchoring, FM anchoring, and a Hutchinson-estimated trace weight, without modifying the backbone.

C3 Our experiments showcase that VaMF can significantly improve the both the convergence efficiency and the performance of the original Mean Flows.

A brief summary of the quantity

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2 Preliminary

Flow-Matching. Let $\mathcal{X} \subset \mathbb{R}^d$ be the space of random variables $\mathbf{x} \in \mathbb{R}^d$ and $\boldsymbol{x}$ a data sample as the realized state. The objective of flow-based generative models is to learn a time-dependent diffeomorphic map $\psi(\boldsymbol{x}, t) : \mathbb{R}^d \times [0, 1] \to \mathbb{R}^d$ from a simple distribution $\mathbf{x}_1 \sim p(\boldsymbol{x}, 1)$ to the target data distribution $\mathbf{x}_0 \sim p(\boldsymbol{x}, 0) = p_{\mathrm{data}}$. Specifically, this is achieved by learning a *time-dependent velocity field* $\boldsymbol{v}(\boldsymbol{x}, t) : \mathbb{R}^d \times [0, 1] \to \mathbb{R}^d$:

$$\boldsymbol{v}(\boldsymbol{x}, t) \triangleq \frac{\mathrm{d}}{\mathrm{d}t}\psi(\boldsymbol{x}, t), \quad \psi(\boldsymbol{x}, 0) = \boldsymbol{x}_0. \tag{1}$$

The map $\psi(\boldsymbol{x}, t)$ induces a *time-dependent probability density path* $p(\boldsymbol{x}, t) : \mathbb{R}^d \times [0, 1] \to \mathbb{R}_{>0}$, with $\int_{\mathcal{X}} p(\boldsymbol{x}, t)\mathrm{d}\boldsymbol{x}, \forall t \in [0, 1]$. A sufficient and necessary condition for a vector field to generate a probability path is the *continuity equation*, which is a partial differential equation (PDE) that writes

$$\frac{\partial}{\partial t} p(\boldsymbol{x}, t) + \mathrm{div}\left(p(\boldsymbol{x}, t)\boldsymbol{v}(\boldsymbol{x}, t)\right) = 0, \tag{2}$$

where $\mathrm{div}(f)$ denotes the divergence of function f. However, the exact *marginal velocity field* $\boldsymbol{v}(\boldsymbol{x}, t)$ is generally *inaccessible*, especially in a high-dimensional sample space. The optimal-transport conditional flow-matching (OT-CFM) [5, 6] alternatively learns a conditional velocity field $\boldsymbol{v}(\boldsymbol{x}, t \mid \boldsymbol{x}_0; \boldsymbol{\theta}) \triangleq \boldsymbol{v}\left((1-t)\boldsymbol{x}_0 + t\boldsymbol{x}_1; \boldsymbol{\theta}\right)$ by minimizing a loss function:

$$\mathcal{L}_{\mathrm{FM}}(\boldsymbol{\theta}) = \mathbb{E}_{t \sim \mathcal{U}(0,1), \boldsymbol{x}_0 \sim p_{\mathrm{data}}, \boldsymbol{x}_1 \sim \mathcal{N}(0, \boldsymbol{I}_d)} \left\| \boldsymbol{v}\left(\boldsymbol{x}, t; \boldsymbol{\theta}\right) - \left(\boldsymbol{x}_1 - \boldsymbol{x}_0\right) \right\|_2^2. \tag{3}$$

Herein, $t \sim \mathcal{U}(0, 1)$ is a scalar sample from the uniform distribution defined on the range of $[0, 1]$ and $\boldsymbol{x}_1 \sim \mathcal{N}(0, \boldsymbol{I}_d)$ is a white noise sample from the standard normal distribution defined in $\mathbb{R}^d$. Meanwhile, there exists a relationship between the marginal and conditional velocity field given by the following lemma:

Lemma 1. *Given vector fields $\boldsymbol{v}(\boldsymbol{x}, t \mid \boldsymbol{x}_0)$ that generate probability paths $p(\boldsymbol{x}, t \mid \boldsymbol{x}_0)$, for any data distribution $\mathbf{x}_0 \sim p(\mathbf{x}_0)$, the marginal velocity field is equal to the expected conditional velocity field*

$$\boldsymbol{v}(\boldsymbol{x}, t) = \mathbb{E}_{\boldsymbol{x}_0 \sim p(\boldsymbol{x}_0, t \mid \boldsymbol{x})}\left[\boldsymbol{v}(\boldsymbol{x}, t \mid \boldsymbol{x}_0)\right]. \tag{4}$$

This is the key to the validity of surrogate loss in equation 3. See Appendix B for its proof.

One-step Mean Flows. The issue with diffusion [11, 13] and flow-matching models [5] is the integration. Evaluating the velocity field multiple times is not only expensive but can also lead to numerical instability. A Mean Flow model [24] instead learns a two-parameter average velocity field $\boldsymbol{u}(\mathbf{x}, r, t)$ to enable efficient sampling. It leverages the identity between the average and marginal velocity field given by

$$(t-r)\boldsymbol{u}(\mathbf{x}, r, t) = \int_{\tau=r}^{t} \boldsymbol{v}(\mathbf{x}, \tau)\mathrm{d}\tau, \tag{5}$$

where r and t are scalars, the start and end time steps along the probability density path. Taking the total derivative with respect to the terminal time step t yields

$$\boldsymbol{u}(\boldsymbol{x}, r, t) + (t-r)\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{u}(\boldsymbol{x}, r, t) = \boldsymbol{v}(\boldsymbol{x}, t), \tag{6}$$

where the total derivative of the average velocity field with respect to t can be expressed as a Jacobi-vector Product (JVP)

$$\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{u}(\mathbf{x}, r, t) = \partial_{\mathbf{x}}\boldsymbol{u} \cdot \boldsymbol{v}(\mathbf{x}, t) + \partial_r \boldsymbol{u} \cdot \frac{\mathrm{d}r}{\mathrm{d}t} + \partial_t \boldsymbol{u} \cdot \frac{\mathrm{d}t}{\mathrm{d}t} = \partial_{\mathbf{x}}\boldsymbol{u} \cdot \boldsymbol{v}(\mathbf{x}, t) + \partial_t \boldsymbol{u} \tag{7}$$

We abuse the notation and express the right-hand side by $\texttt{JVP}(\boldsymbol{u}, (\mathbf{x}, r, t), (\boldsymbol{v}(\mathbf{x}, t), 0, 1))$. In the original MeanFlow [24], the authors replace both the marginal velocity field in the right-hand side of equation 6 and inside the JVP with the conditional velocity field, and derive the MeanFlow loss function given by

$$\mathcal{L}_{\mathrm{MF}}(\boldsymbol{\theta}) = \mathbb{E}_{r,t,\boldsymbol{x}_0,\boldsymbol{x}_1}\left[\left\|\boldsymbol{u}_{\boldsymbol{\theta}} + (t-r)\texttt{sg}\left[\texttt{JVP}(\boldsymbol{u}_{\boldsymbol{\theta}}, (\boldsymbol{z}, r, t), (\boldsymbol{v}_{\mathrm{cond}}, 0, 1))\right] - \boldsymbol{v}_{\mathrm{cond}}\right\|_2^2\right], \tag{8}$$

where $r \sim \mathcal{U}(0, 1)$, $t \sim \mathcal{U}(0, 1)$, $r \leq t$, $\boldsymbol{x}_0 \sim p_{\mathrm{data}}$, $\boldsymbol{x}_1 \sim \mathcal{N}(0, \boldsymbol{I}_d)$, $\boldsymbol{z} = (1-t)\boldsymbol{x}_0 + t\boldsymbol{x}_1$, $\boldsymbol{v}_{\mathrm{cond}} = \boldsymbol{x}_1 - \boldsymbol{x}_0$, and $\texttt{sg}[\cdot]$ is the stop-gradient operator. The authors highlight that stopping gradients in JVP helps eliminate the need for "double backpropagation" and high-order derivatives.

3 Variance-aware Mean Flows

The MeanFlow loss in equation 8 is empirically known to be *non-decreasing* and to suffer from *high-variance gradients* [25, 26]. In this section, we aim to answer the two complementary questions:

> ***RQ1* (Diagnosis): What ist the root cause for the training pathologies in Mean Flows?**
>
> Existing works diagnose MeanFlow's instability from three apparently distinct angles, such as gradient conflict [25] and the curvature bottleneck [27]. Are they symptoms of a single underlying mechanism, and if so, what observable in the gradients does that mechanism affect?

> ***RQ2* (Prescription): Variance-aware loss design.**
>
> Given the identified mechanism, what is the simplest modification that can drive the gradient variance toward its irreducible level?

In the following, section 3.1 answers *RQ1* at the level of the gradients. We apply Reynolds decomposition to reveal both the bias induced by stop-gradient training and the per-step gradient variance governed by the Jacobi factor $\boldsymbol{J} = (t-r)\partial_{\boldsymbol{z}}\boldsymbol{u}_{\boldsymbol{\theta}} - \boldsymbol{I}_d$, and establish a connection between the bias and the *curvature gap* $\Delta = \boldsymbol{u} - \boldsymbol{v}$. Section 3.2 then quantifies the gap in terms of flow acceleration with a Taylor expansion, exposing the material derivative as its physical source of the curvature gap and deriving three design principles. Section 3.3 answers *RQ2* by instantiating the three principles and propose our training framework, *Variance-Aware MeanFlow (VaMF)*, comprising an EMA velocity anchor, a flow-matching anchor, and a closed-form trace weight, all applied on top of the original MSE objective without any architectural change. Full proofs are deferred to Appendix C.

3.1 Bias and Variances in Learning Mean Flows

We begin with investigating the consequences of replacing the inaccessible marginal velocity $\boldsymbol{v}(\boldsymbol{x}, t)$ by the conditional field $\boldsymbol{v}_{\mathrm{cond}} = \boldsymbol{x}_1 - \boldsymbol{x}_0$ in both the JVP tangent and the regression target. We first apply the Reynolds decomposition of the field and let $\boldsymbol{v}(\boldsymbol{x}, t \mid \boldsymbol{x}_0) = \boldsymbol{v}(\boldsymbol{x}, t) + \boldsymbol{v}'$, where $\boldsymbol{v}'$ is a zero-mean fluctuation from the marginal velocity induced by optimal transport between latent $\boldsymbol{x}_1$ and data $\boldsymbol{x}_0$. The regression loss equation 8 expands to

$$\begin{aligned}
\mathcal{L}_{\mathrm{MF}}(\boldsymbol{\theta}) &= \mathbb{E}_{r,t,\boldsymbol{x}_0,\boldsymbol{x}_1}\left[\|\boldsymbol{u}_{\boldsymbol{\theta}} + (t-r)\mathtt{sg}\left[\partial_{\boldsymbol{z}}\boldsymbol{u}_{\boldsymbol{\theta}} \cdot \boldsymbol{v}_{\mathrm{cond}} + \partial_t \boldsymbol{u}_{\boldsymbol{\theta}}\right] - \boldsymbol{v}_{\mathrm{cond}}\|_2^2\right] \\
&= \mathbb{E}_{r,t,\boldsymbol{x}_0,\boldsymbol{x}_1}\left[\|\boldsymbol{u}_{\boldsymbol{\theta}} + (t-r)\mathtt{sg}\left[\partial_{\boldsymbol{z}}\boldsymbol{u}_{\boldsymbol{\theta}} \cdot \boldsymbol{v}(\boldsymbol{x}_t, t) + \partial_t \boldsymbol{u}_{\boldsymbol{\theta}}\right] - \boldsymbol{v}(\boldsymbol{x}_t, t) + \mathtt{sg}[\boldsymbol{J}]\boldsymbol{v}'\|_2^2\right] \\
&= \mathbb{E}_{r,t,\boldsymbol{x}_t,\boldsymbol{x}_1}\left[\mathbb{E}_{\boldsymbol{x}_0|\boldsymbol{x}_t}\left[\left\|\underbrace{\boldsymbol{u}_{\boldsymbol{\theta}} + (t-r)\mathtt{sg}\left[\partial_{\boldsymbol{z}}\boldsymbol{u}_{\boldsymbol{\theta}} \cdot \boldsymbol{v}(\boldsymbol{x}_t, t) + \partial_t \boldsymbol{u}_{\boldsymbol{\theta}}\right] - \boldsymbol{v}(\boldsymbol{x}_t, t)}_{\text{mean-field residual } \boldsymbol{r}_{\boldsymbol{\theta}}^{\mathrm{sg}}(\boldsymbol{x}_t, t)} + \mathtt{sg}[\boldsymbol{J}]\boldsymbol{v}'\right\|_2^2\right]\right].
\end{aligned} \tag{9}$$

Herein, we abuse the notations and let $\boldsymbol{J} \triangleq (t-r)\partial_{\boldsymbol{z}}\boldsymbol{u}_\theta - \boldsymbol{I}_d \in \mathbb{R}^{d\times d}$ denote a Jacobi factor of $\boldsymbol{u}_{\boldsymbol{\theta}}$ given sample $\boldsymbol{z} = \boldsymbol{x}_1 - \boldsymbol{x}_0$.

This decomposition reveals that the original loss for learning Mean Flows is the sum of a deterministic mean-field residual $\boldsymbol{r}_{\boldsymbol{\theta}}^{\mathrm{sg}}(\boldsymbol{x}_t, t)$ and a sample-based Jacobi-velocity-product $\boldsymbol{J}\boldsymbol{v}'$. Importantly, the mean-field residual is exactly the loss we aim to minimize given the identity in equation 6. If we further expand the inner expectation, based on equation 4, we have $\mathbb{E}_{\mathbf{x}_0|\mathbf{x}_t}[\boldsymbol{v}'] = 0$ and the cross term vanishes, resulting in

$$\mathbb{E}_{\boldsymbol{x}_0|\boldsymbol{x}_t}[\ell_{\mathrm{MF}}(\boldsymbol{\theta})] = \mathbb{E}_{\boldsymbol{x}_0|\boldsymbol{x}_t}\left[\|\boldsymbol{r}(\boldsymbol{x}_t, t) + \mathtt{sg}[\boldsymbol{J}\boldsymbol{v}']\|_2^2\right] = \|\boldsymbol{r}_{\boldsymbol{\theta}}(\boldsymbol{x}_t, t)\|_2^2 + \mathtt{sg}[\mathrm{Tr}(\boldsymbol{J}\Sigma_{\boldsymbol{v}'}\boldsymbol{J}^\top)], \tag{10}$$

where $\Sigma_{\boldsymbol{v}'} = \mathrm{Cov}_{\boldsymbol{x}_0|\boldsymbol{x}_t}[\boldsymbol{v}']$ is the *non-zero* covariance of the conditional velocity field and $\ell_{\mathrm{MF}}(\boldsymbol{\theta})$ is the per-step loss. Therefore, replacing marginal velocity fields wih conditional ones gives birth to an irreducible per-sample variance of the conditional fluctuation given by the trace $\mathrm{Tr}(\Sigma_{\boldsymbol{v}'})$, where the matrix $\boldsymbol{J}$ converts it into a per-parameter loss contribution that grows quadratically in the spatial Jacobian. We can prove that the magnitude of the average-velocity Jacobian amplifies the per-step variance of the gradient.

Theorem 2 (Jacobian Variance Amplification.)**.** *The Jacobi factor amplifies the variances of the original MeanFlow loss*

$$\boxed{\mathrm{Var}\left[\nabla_{\boldsymbol{\theta}}\ell_{MF}(\boldsymbol{\theta}) \mid \boldsymbol{x}_t\right] \propto \mathrm{Tr}(\boldsymbol{J}\Sigma_{\boldsymbol{v}'}\boldsymbol{J}^\top)}$$

Theorem 2 explains the instability and high variance observed in training the MeanFlow model. With large intervals $(t-r)$ or large norm of the Jacobi term $\|\partial_{\boldsymbol{x}_t}u_\theta\|_2^2$, the per-step gradient is dominated by the noise term $\boldsymbol{J}\boldsymbol{v}'$, downweighting the mean-field signal from $\boldsymbol{r}_{\boldsymbol{\theta}}^{\mathrm{sg}}(\boldsymbol{x}_t, t)$.

To help reader better understands the existence of such a gap, we visualize the magnitude $\|\boldsymbol{v}'\|$ of this fluctuation on a example 3-Gaussian mixture in fig. 1. It is clear that such a gap is *non-zero*, *increases* along the integration window, and concentrates spatially in a *mode-mixing regions near the latent* where multiple conditional paths overlap.

Crucially, the $\boldsymbol{J}\boldsymbol{v}'$ amplification arises because $\boldsymbol{v}'$ enters through the JVP tangent, where the spatial Jacobian $(t-r)\partial_{\boldsymbol{z}}\boldsymbol{u}_{\boldsymbol{\theta}}$ multiplies it. If the tangent were replaced by *any* deterministic estimate $\hat{\boldsymbol{v}}$, the fluctuation $\boldsymbol{v}'$ would only appear in the regression target, entering the residual linearly rather than through $\boldsymbol{J}$. In this case, the per-step loss reduces to $\|\tilde{\boldsymbol{r}}_{\boldsymbol{\theta}}\|_2^2 + \mathrm{Tr}(\Sigma_{\boldsymbol{v}'})$, where $\tilde{\boldsymbol{r}}_{\boldsymbol{\theta}}$ is a deterministic residual. As a result, the variance drops to the irreducible level $\mathrm{Tr}(\Sigma_{\boldsymbol{v}'})$ without any Jacobian amplification. We higlight this as a theoretical foundation missing from the proposed method in a recent study [26]. Furthermore, since the inner expectation includes the Jacobian amplified trace, we find that the stop-gradient operator creates a *semi-gradient gap*.

Theorem 3 (Semi-Gradient Gap)**.** *The gradients of MeanFlow loss with and without the stop-gradient operator differ by*

$$\boxed{\underbrace{2(t-r)\mathbb{E}_{r,t,\boldsymbol{x}_0,\boldsymbol{x}_1}\left[\nabla_{\boldsymbol{\theta}}\left(\partial_{\boldsymbol{x}_t}\boldsymbol{u}_\theta \cdot \boldsymbol{v}(\boldsymbol{x}_t, t) + \partial_t \boldsymbol{u}_\theta\right) \cdot \boldsymbol{r}_\theta\right]}_{\textit{mean-field gradient difference}} + \underbrace{\mathbb{E}_{r,t,\boldsymbol{x}_0,\boldsymbol{x}_1}\left[\nabla_{\boldsymbol{\theta}}\,\mathrm{Tr}(\boldsymbol{J}\Sigma_{\boldsymbol{v}'}\boldsymbol{J}^\top)\right]}_{\textit{variance-driven gradient difference}}}$$

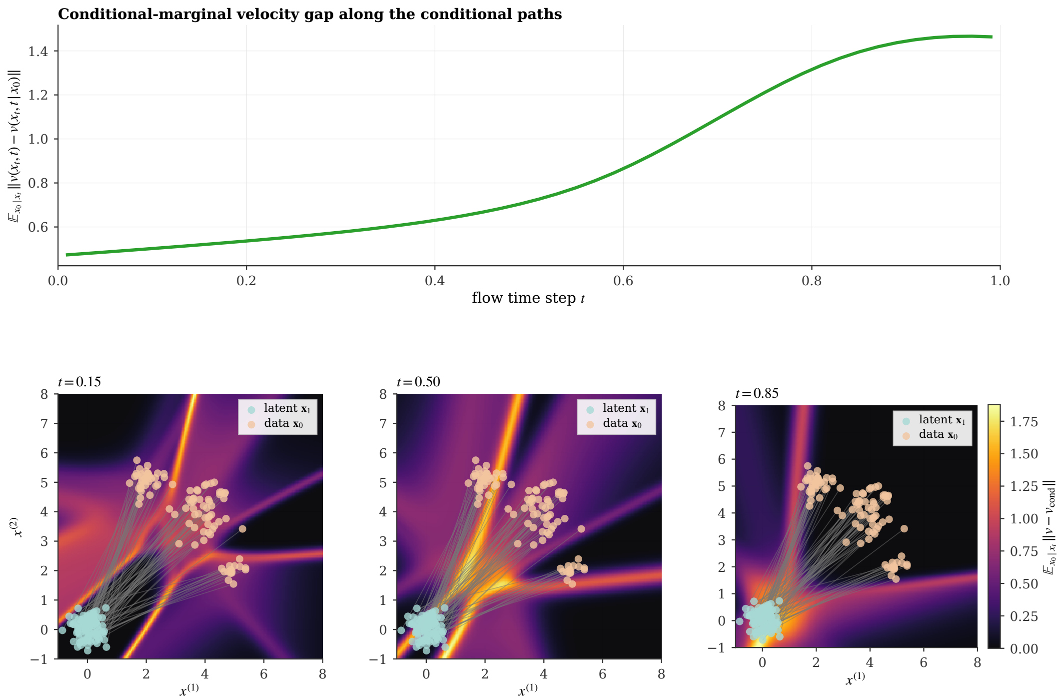

Figure 1: Conditional-marginal velocity gap on a 3-mode mixture. **Top:** the expected per-sample fluctuation $\mathbb{E}_{\boldsymbol{x}_0|\boldsymbol{x}_t}\|\boldsymbol{v}(\boldsymbol{x}_t,t)-\boldsymbol{v}(\boldsymbol{x}_t,t\mid\boldsymbol{x}_0)\|$, evaluated along the conditional paths, grows monotonically with t as conditional trajectories disperse, with the steepest rise where $\boldsymbol{x}_t$ traverses the mode-mixing region. **Bottom:** the spatial heatmap of the same quantity at three timesteps confirms that the gap is heterogeneous and concentrates in the mode-mixing regions where multiple conditional paths overlap. This is the empirical realization of $\Sigma_{\boldsymbol{v}'}$ in equation 9, which the Jacobi factor $\boldsymbol{J}$ amplifies into the gradient variance of theorem 2; the same regions are where the curvature gap of theorem 4 is largest, motivating the trace-weight reweighting of equation 16.

Herein, the stop-gradient operator eliminates the trace term from the expected gradient and $\boldsymbol{r}_{\boldsymbol{\theta}}\to\mathbf{0}$ at convergence. Therefore, the variance-driven term dominates the semi-gradient gap near convergence. $\nabla_{\boldsymbol{\theta}}\operatorname{Tr}(\boldsymbol{J}\Sigma_{\boldsymbol{v}'}\boldsymbol{J}^{\top})$ forces the system to minimize $\|(t-r)\partial_{\mathbf{x}}\boldsymbol{u}_{\boldsymbol{\theta}}-\boldsymbol{I}_d\|$, essentially pushes $\partial_{\mathbf{x}}\boldsymbol{u}_{\boldsymbol{\theta}}\to\frac{1}{t-r}\boldsymbol{I}_d$ to minimize the magnitude of $\boldsymbol{J}$. On the contrary, stop-gradient operator blinds variance-driven gradient, and as a result, $\operatorname{Tr}(\boldsymbol{J}\Sigma_{\boldsymbol{v}'}\boldsymbol{J}^{\top})$ can potentially increase faster than the magnitude of decrease in $\|\boldsymbol{r}_{\boldsymbol{\theta}}\|_2^2$, leading to the non-decreasing loss.

Both failure modes—Jacobian variance amplification (theorem 2) and the unobserved variance growth from the semi-gradient gap (theorem 3)—share a common root. Define the *curvature gap*

$$\Delta(\boldsymbol{x}_t,r,t):=\boldsymbol{u}(\boldsymbol{x}_t,r,t)-\boldsymbol{v}(\boldsymbol{x}_t,t),\tag{11}$$

the difference between the average and instantaneous velocity fields. Rearranging the MeanFlow identity equation 6 gives $(t-r)\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{u}=-\Delta$. The mean-field residual in equation 9 can therefore be rewritten as $\boldsymbol{r}_{\boldsymbol{\theta}}^{\mathrm{sg}}=\Delta_{\boldsymbol{\theta}}+(t-r)\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{u}_{\boldsymbol{\theta}}$, where $\Delta_{\boldsymbol{\theta}}:=\boldsymbol{u}_{\boldsymbol{\theta}}-\boldsymbol{v}$. At convergence the total derivative must exactly cancel the curvature gap ($\Delta_{\boldsymbol{\theta}}+(-\Delta_{\boldsymbol{\theta}})=\mathbf{0}$), so the training signal $\boldsymbol{r}_{\boldsymbol{\theta}}^{\mathrm{sg}}$ is a small difference of two quantities whose magnitudes scale with $\|\Delta\|$. When $\|\Delta\|$ is large, this cancellation is delicate: the gradient signal from $\boldsymbol{r}_{\boldsymbol{\theta}}$ is weak relative to the noise $\boldsymbol{J}\boldsymbol{v}'$ from theorem 2, and the semi-gradient gap from theorem 3 prevents the optimizer from controlling the resulting variance growth. This raises a natural question: how large is Δ in practice, and what governs its magnitude?

3.2 Flow Acceleration

The curvature gap $\Delta(\boldsymbol{x}_t,r,t)$ defined in equation 11 controls the severity of both Jacobian variance amplification and the semi-gradient gap. We now quantify Δ using flow acceleration, revealing its dependence on the material derivative of the velocity field.

Theorem 4 (Flow Acceleration Expansion of the Expectation Gap)**.** *Let $\boldsymbol{v}(\mathbf{x}, t) \in C^2$ be twice continouously differentiable. The expectation gap admits a Taylor series at $\boldsymbol{x}_t$:*

$$\Delta(\boldsymbol{x}_t, r, t) = -\frac{(t-r)}{2}\frac{\mathrm{D}v}{\mathrm{D}t}(\boldsymbol{x}_t, t) + \frac{(t-r)^2}{6}\frac{\mathrm{D}^2 v}{\mathrm{D}t^2}(\boldsymbol{x}_t, t) + \mathcal{R}_3(\boldsymbol{x}_t, r, t), \tag{12}$$

where $\frac{\mathrm{D}v}{\mathrm{D}t}(\mathbf{x}, t) \triangleq \partial_t \boldsymbol{v}(\mathbf{x}, t) + (\nabla_{\mathbf{x}} \boldsymbol{v}(\mathbf{x}, t)) \cdot \boldsymbol{v}(\mathbf{x}, t)$ is the material dervative (a.k.a. flow acceleration) and $\mathcal{R}_3(\mathbf{x}, r, t)$ is the third-order remainder of the Taylor series.

Herein, it is reasonable to assume $\boldsymbol{v}(\mathbf{x}, t) \in C^2$ since existing work primarily uses a U-Net [28, 12] or transformer backbone with twice continuously differentiable activation functions (*e.g., the GELU activation [29]*) to learn the velocity field. Theorem 4 establishes a theoretical perspective that bridges the points discussed in the existing works. Specifically, methods with rectified flows [30] straighten the marginal ODE trajectory, thereby enforcing near-zero flow accelerations. By theorem 4, this pushes $\Delta(\mathbf{x}, r, t) \to 0$ and eliminates the root cause for training instability.

Meanwhile, methods that employ progressive scheduling [26] address the problem by initially training with a small interval $r \to t$. According to theorem 4, the expectation gap grows at a rate factor $\frac{(t-r)}{2}\left\|\frac{\mathrm{D}}{\mathrm{D}t}(\mathbf{x}, r, t)\right\|$ and hence it is reasonable to constrain the maximum interval during training by bounding this factor. The expansion enables us to establish a Lipschitz bound on the expectation gap.

In addition, the relationship between the expectation gap and the flow acceleration reveals a challenge in the intermediate time step t, where the flow acceleration maximizes due to the intersection of conditional trajectories. The issue is exacerbated for high-resolution, multi-modal data, where marginal velocity field changes occur rapidly.

Together with the variance analysis in section 3.1, the material derivative expansion yields three design principles for stabilizing MeanFlow training:

D1 Eliminate variance amplification. As shown above, $\boldsymbol{v}'$ enters through the JVP tangent and is amplified by $\boldsymbol{J}$ (theorem 2). Replacing the stochastic tangent with a deterministic estimate moves $\boldsymbol{v}'$ outside the JVP, reducing variance to the irreducible level $\mathrm{Tr}(\Sigma_{\boldsymbol{v}'})$.

D2 Correct tangent bias. Any deterministic tangent that differs from $\boldsymbol{v}(\boldsymbol{x}_t, t)$ introduces a compound-prediction bias. The semi-gradient gap (theorem 3) blinds the optimizer to this bias, so auxiliary boundary supervision is needed.

D3 Adapt to local curvature. Since $\|\Delta\| \propto (t-r)\left\|\frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t}\right\|$ by theorem 4, per-sample difficulty varies with both interval length and flow acceleration. Optimal weighting should downweight high-curvature, long-interval samples where the training signal is weakest.

3.3 Variance-Aware MeanFlow

We now instantiate these three design principles into a concrete training framework—Variance-Aware MeanFlow (VaMF)—that addresses all identified failure modes without modifying the backbone architecture.

3.3.1 Velocity Anchoring (D1)

To eliminate the Jacobian variance amplification identified in theorem 2, we replace the stochastic JVP tangent $\boldsymbol{v}_{\mathrm{cond}}$ with a deterministic estimate. Following iMF [26], we use a Polyak averaging [31] (*a.k.a. exponential moving average (EMA)*) [32] of the model parameters as the tangent. Specifically, the new loss writes

$$\mathcal{L}_{\mathrm{MF}}^{\mathrm{EMA}}(\boldsymbol{\theta}) = \mathbb{E}_{r,t,\boldsymbol{x}_0,\boldsymbol{x}_1}\left[\left\|\boldsymbol{u}_{\boldsymbol{\theta}} + (t-r)\mathtt{sg}\left[\mathtt{JVP}(\boldsymbol{u}_{\boldsymbol{\theta}}, (\boldsymbol{z}, r, t), (\boldsymbol{u}_{\bar{\boldsymbol{\theta}},t}, 0, 1))\right] - \boldsymbol{v}_{\mathrm{cond}}\right\|_2^2\right], \tag{13}$$

where $\boldsymbol{u}_{\bar{\boldsymbol{\theta}},t} = \boldsymbol{u}(\boldsymbol{x}_t, t, t)$ and $\bar{\boldsymbol{\theta}}$ is updated by $\bar{\boldsymbol{\theta}} \leftarrow \mu\bar{\boldsymbol{\theta}} + (1-\mu)\boldsymbol{\theta}$ with $\mu \in [0, 1)$.

Remark 5. The Polyak averaging slowly evolves $\boldsymbol{u}_{\bar{\boldsymbol{\theta}}}$ to the online model $\boldsymbol{u}_{\boldsymbol{\theta}}$. In the early stage of the training, both $\boldsymbol{u}_{\boldsymbol{\theta}}$ and $\boldsymbol{u}_{\bar{\boldsymbol{\theta}}}$ are poor estimators. Still, the anchoring provides a direct supervision that rapidly improves $\boldsymbol{u}_{\boldsymbol{\theta}}$, and hence indirectly improves $\boldsymbol{u}_{\bar{\boldsymbol{\theta}}}$.

3.3.2 The VA-MeanFlow Objective (D2)

The EMA tangent solves D1 but introduces a transient bias when $\boldsymbol{u}_{\bar{\boldsymbol{\theta}}}(\boldsymbol{x}_t, t, t) \neq \boldsymbol{v}(\boldsymbol{x}_t, t)$, leaving D2 unaddressed. To correct this, we add a *flow matching anchor* that provides direct velocity supervision near the boundary $r \to t$:

$$\mathcal{L}_{\mathrm{FM}}(\boldsymbol{\theta}) = \mathbb{E}_{t,\boldsymbol{x}_0,\boldsymbol{x}_1}\left[\left\|\boldsymbol{u}_{\boldsymbol{\theta}}(\boldsymbol{x}_t, t-\delta, t) - \boldsymbol{v}_{\mathrm{cond}}\right\|_2^2\right], \tag{14}$$

where $\delta \sim \mathcal{U}[\delta_{\min}, \delta_{\max}]$ with $\delta_{\max} \ll 1$. For small δ, the average velocity over $[t-\delta, t]$ approximates the instantaneous velocity, so equation 14 is asymptotically equivalent to equation 3 without a separate network head. The full **VA-MeanFlow** objective is:

$$\mathcal{L}_{\text{VA-MF}}(\boldsymbol{\theta}) = \mathcal{L}_{\mathrm{MF}}^{\mathrm{EMA}}(\boldsymbol{\theta}) + \lambda\, \mathcal{L}_{\mathrm{FM}}(\boldsymbol{\theta}), \tag{15}$$

where $\lambda > 0$ balances the two terms. The following propositions characterize the gradient structure.

Proposition 6 (Gradient of VA-MF Loss). *The gradient of $\mathcal{L}_{MF}^{EMA}$ takes the form $\nabla_{\boldsymbol{\theta}}\mathcal{L}_{MF}^{EMA} = 2\,\mathbb{E}\big[\nabla_{\boldsymbol{\theta}}\boldsymbol{u}_{\boldsymbol{\theta}}(\boldsymbol{x}_t, r, t) \cdot \tilde{\boldsymbol{r}}^{EMA}\big]$, where $\tilde{\boldsymbol{r}}^{EMA} := V_{\boldsymbol{\theta}}^{EMA} - \boldsymbol{v}(\boldsymbol{x}_t, t)$ is the mean-field residual under the EMA tangent. The conditional velocity variance does not contribute.*

Proposition 7 (Bias-Variance Tradeoff). *The EMA mean-field residual decomposes as $\tilde{\boldsymbol{r}}^{EMA} = \boldsymbol{r}_{\boldsymbol{\theta}} + (t-r)\,\partial_{\boldsymbol{z}}\boldsymbol{u}_{\boldsymbol{\theta}}(\boldsymbol{u}_{\bar{\boldsymbol{\theta}}}(\boldsymbol{x}_t, t, t) - \boldsymbol{v}(\boldsymbol{x}_t, t))$, where $\boldsymbol{r}_{\boldsymbol{\theta}}$ is the true MeanFlow residual from equation 9. At convergence, $\boldsymbol{r}_{\boldsymbol{\theta}} \to \mathbf{0}$ and the FM anchor drives $\boldsymbol{u}_{\bar{\boldsymbol{\theta}}}(\boldsymbol{x}_t, t, t) \to \boldsymbol{v}(\boldsymbol{x}_t, t)$, giving $\tilde{\boldsymbol{r}}^{EMA} \to \mathbf{0}$.*

Proposition 6 confirms that the VA-MF gradient has a first-order structure with no Jacobian-mediated variance amplification (cf. theorem 2). Proposition 7 shows the transient bias is bounded by $(t-r)\|\partial_{\boldsymbol{z}}\boldsymbol{u}_{\boldsymbol{\theta}}\|\|\boldsymbol{u}_{\bar{\boldsymbol{\theta}}}(\boldsymbol{x}_t, t, t) - \boldsymbol{v}(\boldsymbol{x}_t, t)\|$ and shrinks as the EMA velocity improves.

3.3.3 Trace Weight (D3)

With D1 and D2 resolved, the per-sample magnitude of the EMA loss in equation 13 is still heterogeneous across $(\boldsymbol{x}_t, r, t)$: the inner expectation contains the curvature-amplified noise $\mathrm{Tr}(\boldsymbol{J}\Sigma_{\boldsymbol{v}'}\boldsymbol{J}^\top)$ from theorem 2. Even though the EMA tangent removes this term from the *gradient*, samples with large $\|\boldsymbol{J}\|$ still produce loss values that dwarf samples in flat, low-curvature regions, biasing the empirical-risk estimator toward the noisy regime. Design principle D3 addresses this by *reweighting* each sample to equalize its contribution to the expected loss.

BLUE matrix weight. Treating the conditional fluctuation $\boldsymbol{v}'$ as Gaussian noise, the Best Linear Unbiased Estimator (BLUE) of the mean-field residual $\tilde{\boldsymbol{r}}^{\mathrm{EMA}}$ uses the inverse-covariance matrix weight $(\boldsymbol{J}\Sigma_{\boldsymbol{v}'}\boldsymbol{J}^\top)^{-1}$. Two practical obstacles rule out the matrix form: **(i)** the per-sample $d \times d$ matrix inversion costs $\mathcal{O}(d^3)$ and is intractable for high-dimensional images or latents; **(ii)** the matrix is ill-conditioned whenever $(t-r)\partial_{\boldsymbol{z}}\boldsymbol{u}_{\boldsymbol{\theta}} \approx \boldsymbol{I}_d$, where $\boldsymbol{J} \approx \mathbf{0}$ pushes the inverse to $+\infty$.

Scalar surrogate. Under the isotropic-noise approximation $\Sigma_{\boldsymbol{v}'} \approx \sigma_t^2 \boldsymbol{I}_d$, both pathologies are absorbed by replacing the matrix weight with a single-scalar trace-based surrogate. Because $\mathrm{Tr}(\boldsymbol{J}\Sigma_{\boldsymbol{v}'}\boldsymbol{J}^\top) = \sigma_t^2\,\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)$, we define the per-sample *trace weight*

$$w(\boldsymbol{x}_t, r, t) \;:=\; \frac{1}{1 + \sigma_t^2\, \mathtt{sg}[\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)]\,/d}\,, \tag{16}$$

where the constant d in the denominator of the curvature term is the per-sample dimensionality and the additive 1 provides the numerical stability that the BLUE inverse lacks: $w \to 1$ as $\boldsymbol{J} \to \mathbf{0}$ (low curvature, no reweighting) and $w \to d/(\sigma_t^2\,\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top))$ as the curvature term dominates (recovering the proportionality of the BLUE-scalar limit). The stop-gradient on $\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)$ ensures the weight does not contribute to the parameter update, so w acts purely as a per-sample reweighting.

Hutchinson estimator. Direct evaluation of $\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)$ requires d Jacobian-vector products and is intractable for image-scale data. We instead use a single Rademacher (or Gaussian) probe $\boldsymbol{z} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{I}_d)$ and the unbiased estimator [33]

$$\widehat{\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)} \;=\; \left\|\boldsymbol{J}\boldsymbol{z}\right\|_2^2 \;=\; \left\|(t-r)\,\partial_{\boldsymbol{z}}\boldsymbol{u}_{\boldsymbol{\theta}}\,\boldsymbol{z} - \boldsymbol{z}\right\|_2^2, \tag{17}$$

which costs a single JVP per step. Multiple probes can be averaged when extra accuracy is needed; $K=1$ is sufficient for the experiments in this paper.

Algorithm 1 Variance-Aware MeanFlow Training

Require: Dataset $\mathcal{D}$, EMA decay μ, anchor weight λ, anchor interval $[\delta_{\min}, \delta_{\max}]$, schedule σ_t, probes K

Initialize $\bar{\boldsymbol{\theta}} \leftarrow \boldsymbol{\theta}$

for $k = 1, 2, \ldots$ **do**

  Sample $(\boldsymbol{x}_0, \boldsymbol{x}_1) \sim \mathcal{D} \times \mathcal{N}(\boldsymbol{0}, \boldsymbol{I}_d)$, $t \sim \mathcal{U}[0,1]$, $r \sim \mathcal{U}[0,t]$, $\delta \sim \mathcal{U}[\delta_{\min}, \delta_{\max}]$

  $\boldsymbol{x}_t \leftarrow (1-t)\boldsymbol{x}_0 + t\,\boldsymbol{x}_1$, $\quad \boldsymbol{v}_{\text{cond}} \leftarrow \boldsymbol{x}_1 - \boldsymbol{x}_0$

  $\boldsymbol{v}_{\text{tang}} \leftarrow \mathtt{sg}[\boldsymbol{u}_{\bar{\boldsymbol{\theta}}}(\boldsymbol{x}_t, t, t)]$ ▷ EMA tangent (D1)

  $V_{\boldsymbol{\theta}} \leftarrow \boldsymbol{u}_{\boldsymbol{\theta}}(\boldsymbol{x}_t, r, t) + (t-r)\,\mathtt{sg}[\mathtt{JVP}(\boldsymbol{u}_{\boldsymbol{\theta}}, (\boldsymbol{x}_t, r, t), (\boldsymbol{v}_{\text{tang}}, 0, 1))]$

  Sample K probes $\{\boldsymbol{z}_i\}_{i=1}^K \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{I}_d)$

  $\widehat{\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)} \leftarrow \frac{1}{K}\sum_{i=1}^K \left\|(t-r)\,\partial_{\boldsymbol{z}}\boldsymbol{u}_{\boldsymbol{\theta}}(\boldsymbol{x}_t, r, t)\,\boldsymbol{z}_i - \boldsymbol{z}_i\right\|_2^2$

  $w \leftarrow 1 \,/\, \left(1 + \sigma_t^2\,\mathtt{sg}[\widehat{\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)}]/d\right)$ ▷ trace weight (D3)

  $\ell_{\text{MF}} \leftarrow w \cdot \|V_{\boldsymbol{\theta}} - \boldsymbol{v}_{\text{cond}}\|_2^2$

  $\ell_{\text{FM}} \leftarrow \|\boldsymbol{u}_{\boldsymbol{\theta}}(\boldsymbol{x}_t, t-\delta, t) - \boldsymbol{v}_{\text{cond}}\|_2^2$ ▷ FM anchor (D2)

  $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \eta\,\nabla_{\boldsymbol{\theta}}(\ell_{\text{MF}} + \lambda\,\ell_{\text{FM}})$

  $\bar{\boldsymbol{\theta}} \leftarrow \mu\bar{\boldsymbol{\theta}} + (1-\mu)\boldsymbol{\theta}$ ▷ EMA update

end for

return $\boldsymbol{\theta}$ ▷ Generate: $\hat{\boldsymbol{x}}_0 = \boldsymbol{x}_1 - \boldsymbol{u}_{\boldsymbol{\theta}}(\boldsymbol{x}_1, 0, 1)$

Schedule σ_t. The scaling factor σ_t in equation 16 absorbs the conditional-velocity variance scale and re-introduces a controllable schedule along the time axis. We consider three options: $\sigma_t = 1$ (no schedule; the curvature term is the only modulation), $\sigma_t = t^2$ (default; matches the variance growth along linear interpolation paths), and a small learned MLP $\sigma_t = \mathtt{NN}_\phi(t)$. We use $\sigma_t = t^2$ unless stated otherwise; the ablation in Appendix D confirms it is the most robust across our toy datasets.

VaMF objective. Combining the EMA tangent equation 13, FM anchor equation 14, and trace weight equation 16 yields the full **Variance-Aware MeanFlow** loss

$$\mathcal{L}_{\text{VaMF}}(\boldsymbol{\theta}) \;=\; \mathbb{E}_{r,t,\boldsymbol{x}_0,\boldsymbol{x}_1}\Big[w(\boldsymbol{x}_t, r, t)\,\big\|\tilde{\boldsymbol{r}}_{\boldsymbol{\theta}}^{\text{EMA}}\big\|_2^2\Big] \;+\; \lambda\,\mathcal{L}_{\text{FM}}(\boldsymbol{\theta}), \tag{18}$$

which preserves the original MeanFlow MSE form—no additional network heads, no NLL.

Proposition 8 (Trace-Weight Properties)**.** *Under the trace-weight loss equation 18 and the isotropic approximation $\Sigma_{\boldsymbol{v}'} \approx \sigma_t^2 \boldsymbol{I}_d$: **(a)** The mean-parameter gradient is the trace-weighted form of proposition 6 with w as a stop-gradient scalar, so the BLUE-style normalization does not enter the parameter update. **(b)** The expected per-sample contribution to the loss decomposes into a deterministic signal $w\,\|\tilde{\boldsymbol{r}}^{EMA}\|_2^2$ and a constant noise floor $w\,\sigma_t^2\,\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top) = d\,\sigma_t^2\,\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)/(d+\sigma_t^2\,\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)) \le d$, eliminating the unbounded curvature growth from theorem 2. **(c)** w is monotonically decreasing in $\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)$ and, by theorem 4, in the squared interval length $(t-r)^2\|\frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t}\|^2$, so high-curvature long-interval samples are automatically downweighted—subsuming inverse-variance weighting, progressive scheduling, and acceleration-aware interval sampling in a single per-sample scalar.*

The full training procedure is given in Algorithm 1; the connection between the trace weight and the acceleration-to-noise ratio ANR $= (t-r)\|\frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t}\|/\sqrt{\mathrm{Tr}(\Sigma_{\boldsymbol{v}'})}$ is formalized in Appendix D. At generation time only $\boldsymbol{u}_{\boldsymbol{\theta}}(\boldsymbol{x}_1, 0, 1)$ is needed; the trace-weight machinery exists purely to stabilize training.

4 Related Work

One-Step Generation. Reducing the number of function evaluations to one is a central goal. DDIM [17] first enabled deterministic sampling with larger step sizes from diffusion models. Consistency models [21, 34, 22] enforce that all points on a PF-ODE trajectory map to the same endpoint, with extensions to trajectory consistency [35] and easy fine-tuning [36]. Distillation-based methods [18, 19, 20, 37] achieve strong FIDs [38] on standard benchmarks [39] but require a pre-trained teacher. InstaFlow [40] combines rectified flow with distillation for text-to-image one-step generation. Flow map matching [23] and shortcut models [41] provide principled frameworks for two-time maps. Recent work unifies consistency and flow matching [42, 43], and FACM [44] anchors consistency training with a flow matching loss—a mechanism related to our FM anchor, though within a different training framework.

MeanFlow and Variants. MeanFlow [24] learns the average velocity field via a total-derivative identity, enabling distillation-free one-step generation. AlphaFlow [25] identifies gradient conflict between the flow matching and total consistency components, proposing α-scheduling to manage the tradeoff. Improved MeanFlow [26] traces instability to the stochastic JVP tangent and introduces a separate velocity head with stop-gradient. Re-MeanFlow [27] addresses trajectory curvature through a reflow pre-processing step. Terminal Velocity Matching (TVM) [45] sidesteps the spatial Jacobian entirely by differentiating w.r.t. the terminal time, and Functional Mean Flows [46] independently establishes the marginal-conditional gap for average velocities in a Hilbert space setting. Physics-informed distillation [47] derives a related PDE-based loss for learning flow maps. Our curvature-variance principle unifies these diagnostics—our method addresses all failure modes without architectural changes, separate heads, or pre-training.

Variance Reduction in Flow Training. Several concurrent works address variance in flow-based training. Stable Velocity [48] reveals a two-regime variance structure in flow matching and proposes composite conditioning; TPC [49] reduces variance through temporal pair coupling; and preconditioned flow matching [50] addresses ill-conditioning via anisotropic preconditioning. Bertrand et al. [51] argue that stochastic target variance is negligible for standard flow matching—consistent with our analysis, since the problematic amplification is specific to the MeanFlow JVP, not the flow matching loss itself. Rectified flows [30, 52] reduce trajectory curvature by iterative straightening, which by theorem 4 drives $\frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t} \to 0$.

5 Experiments

We evaluate VaMF on three fronts: **(i)** a 2-D toy benchmark spanning four standard distributions, where we measure both sliced Wasserstein-1 and the variance-amplification diagnostics that motivate the trace weight; **(ii)** a high-dimensional dense-Gaussian-mixture scaling study in which we sweep the data dimensionality from $d{=}2$ up to $d{=}64$ and ablate the schedule σ_t in equation 16; and **(iii)** a class-conditional latent-space training of DiT-B/4 on ImageNet-256×256 against the matched baseline MeanFlow recipe of [24]. All toy and DGMM experiments use a 3-layer MLP with 128 hidden units, $200{,}000$ optimization steps, and three random seeds $\{42, 0, 1\}$; the DiT experiment uses a single seed and $300{,}000$ steps, matching the official recipe of [24]. Code, configurations, and result JSONs are released with the paper.

5.1 Variance Amplification and the Trace Term

We first probe the per-sample loss variance and Jacobian-deviation norm on a frozen checkpoint trained on the eight-Gaussians dataset. Figure 2 plots the variance ratio $\mathrm{Var}[\mathcal{L}_{\text{stoch}}]/\mathrm{Var}[\mathcal{L}_{\text{det}}]$ as a function of t (panel a), the Frobenius norm $\|(t{-}r)J - I\|_F$ that drives this amplification (panel b), and the resulting effect on swiss_roll generation (panel c). The variance ratio peaks at $\sim 259\times$ near $t \approx 0.4$, in direct empirical agreement with theorem 2; the Jacobian-deviation norm peaks in the same regime, identifying it as the immediate source of the amplification. The trace-weight reweighting reduces SW_1 on swiss_roll from 0.072 (MeanFlow) to 0.044 (VaMF-TW) at matched seed and step count.

5.2 Toy Benchmark

Table 1 reports both SW_1 and SW_2 on the four 2-D toy datasets, averaged over three random seeds. VaMF-TW achieves the best score on every dataset under both metrics, with relative SW_1 improvements of -23.4% on `checkerboard`, -13.3% on `eight_gaussians`, -22.6% on `two_moons`, and -54.0% on `swiss_roll`. The SW_2 improvements are systematically larger (-25.2%, -3.6%, -47.5%, and -66.0% respectively): the squared-distance metric weights configurations with isolated outliers more heavily, so the larger SW_2 gains are direct evidence that VaMF-TW reduces the per-sample tail of the residual distribution—exactly the variance-amplification regime that the trace weight was designed to suppress. VaMF-L_2, which keeps the EMA tangent and FM anchor

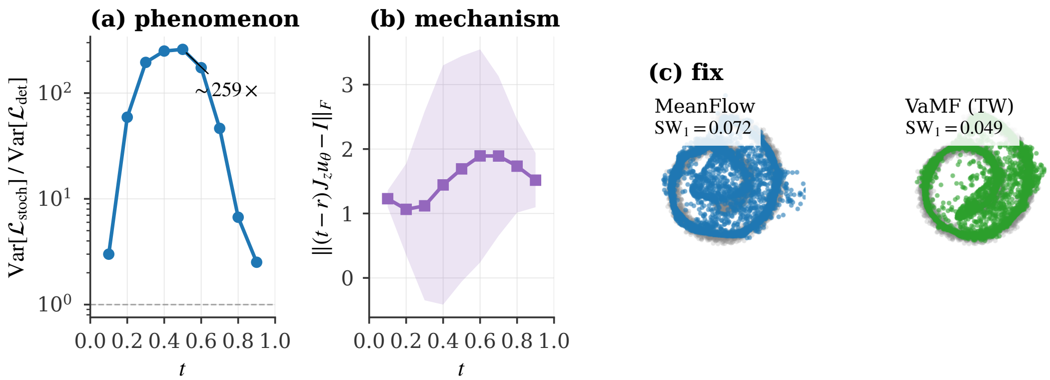

Figure 2: Variance amplification in vanilla MeanFlow and the trace-weight fix. **(a)** Variance ratio $\mathrm{Var}[\mathcal{L}_{\mathrm{stoch}}]/\mathrm{Var}[\mathcal{L}_{\mathrm{det}}]$ peaks at $\sim 259\times$ near $t \approx 0.4$, in direct empirical agreement with theorem 2. **(b)** Frobenius norm $\|(t-r)J - I\|_F$ peaks in the same regime, identifying the linearised flow Jacobian's deviation from identity as the immediate source of (a). **(c)** The trace-weight reweighting reduces SW_1 on swiss_roll from 0.072 (MeanFlow) to 0.044 (VaMF-TW) at matched seed and step count.

Table 1: Sliced Wasserstein distance (lower is better) on the 2-D toy benchmark, averaged over three seeds $\{42, 0, 1\}$ at 200k steps. SW_p is the sliced Wasserstein-p distance evaluated on 4096 samples with 500 random projections; both metrics use the *same* projection key per evaluation for variance-reduced paired estimates. VaMF-TW uses $\sigma_t = t^2$ and a single Hutchinson probe.

Dataset	Metric	MeanFlow	VaMF-L$_2$	VaMF-TW
`checkerboard`	SW$_1$	0.175 ± 0.036	0.165 ± 0.034	$\mathbf{0.134 \pm 0.029}$
	SW$_2$	0.234 ± 0.042	0.211 ± 0.031	$\mathbf{0.175 \pm 0.046}$
`eight_gaussians`	SW$_1$	0.098 ± 0.016	0.087 ± 0.024	$\mathbf{0.085 \pm 0.010}$
	SW$_2$	0.151 ± 0.022	0.161 ± 0.053	$\mathbf{0.146 \pm 0.023}$
`two_moons`	SW$_1$	0.093 ± 0.028	0.087 ± 0.026	$\mathbf{0.072 \pm 0.019}$
	SW$_2$	0.177 ± 0.068	0.148 ± 0.060	$\mathbf{0.093 \pm 0.025}$
`swiss_roll`	SW$_1$	0.098 ± 0.058	0.048 ± 0.008	$\mathbf{0.045 \pm 0.003}$
	SW$_2$	0.177 ± 0.151	0.082 ± 0.029	$\mathbf{0.060 \pm 0.007}$

(D1, D2) but drops the trace weight, recovers a partial improvement over vanilla MeanFlow but trails VaMF-TW on every dataset, confirming that the curvature-aware reweighting (D3) is the load-bearing piece. The qualitative picture is consistent with the SW numbers: Figure 3 shows the full samples grid, and Figure 4 shows the SW$_1$ trajectory across training, where vanilla MeanFlow on `swiss_roll` exhibits the high-variance plateau predicted by theorem 2 while both VaMF variants converge smoothly.

5.3 High-Dimensional Scaling and the σ_t Schedule

We sweep the data dimensionality on the dense Gaussian mixture (DGMM) defined in our toy framework—64 overlapping isotropic components on a unit sphere of radius 1.5 in $\mathbb{R}^d$, $d \in \{2, 4, 8, 16, 32, 64\}$—which is constructed so that variance amplification dominates the optimization landscape. Table 2 and Figure 5 report SW$_1$ at convergence (seed 42, single run per cell). VaMF-TW is the best or tied-best for every dimension $d \geq 4$, and the gap to MeanFlow widens with d (e.g. -16.5% at d=64 versus $+20.1\%$ at d=2, where the curvature term is too small to matter). Table 3 ablates the schedule σ_t in equation 16: $\sigma_t = t^2$ is best or tied-best on five of six dimensions, justifying it as our default; $\sigma_t = 1$ is competitive in low dimension but loses at d=2, where the curvature term lacks the time-axis attenuation; the learned MLP variant tracks the quadratic schedule but offers no consistent improvement, suggesting the closed-form schedule is near-optimal in this regime.

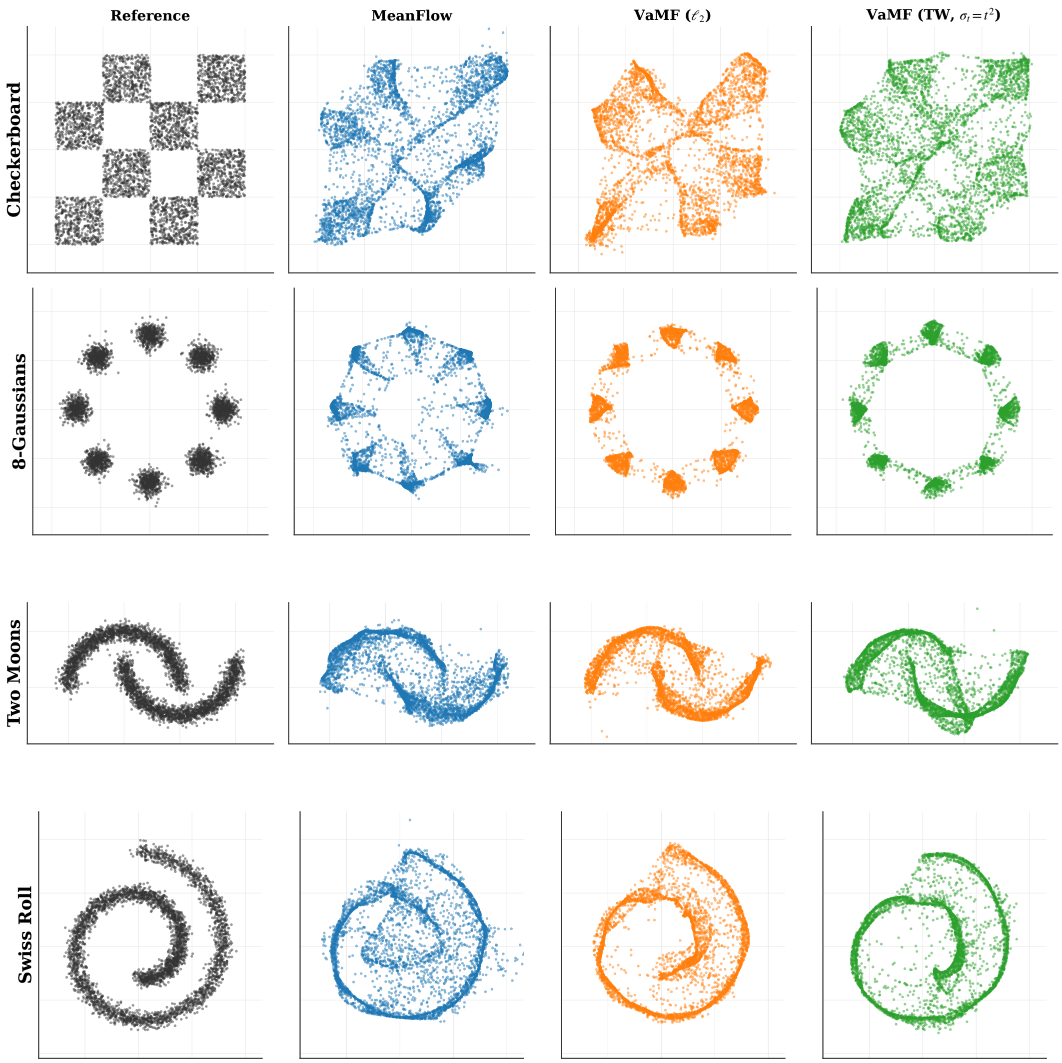

Figure 3: Reference distribution and final-step samples on the four 2-D toy datasets at seed 42. VaMF-TW (third column from the right) recovers the structural details—spirals, sharp corners, mode separation—that vanilla MeanFlow blurs out, especially on `checkerboard` and `swiss_roll`.

Table 2: DGMM scaling (single seed, 200k steps): SW_1 at convergence as the dimension d grows. The trace weight in VaMF-TW becomes more useful with increasing d, while VaMF-L_2 tracks MeanFlow.

d	2	4	8	16	32	64
MeanFlow	**0.097**	0.091	0.078	**0.050**	0.035	0.036
VaMF-L_2	0.089	0.090	0.079	**0.049**	**0.034**	0.031
VaMF-TW	0.117	**0.089**	**0.077**	0.051	**0.035**	**0.030**

5.4 Latent-Space Training of DiT-B/4 on ImageNet-256

We test whether the curvature-variance principle scales beyond toys by training DiT-B/4 [53] on ImageNet-256×256 in the latent space of a frozen Stable Diffusion VAE, following the recipe of [24] (logit-normal (r, t) sampling with mean -0.4 and stddev 1, overlap rate 0.75, AdamW with constant learning rate 10^{-4}, EMA decay 0.9999, batch size 256 across 32 TPU v4 chips, 300,000 steps). Both runs share identical hyperparameters and code paths; only the loss differs (vanilla MeanFlow MSE versus the trace-weighted MSE in equation 18 with $\sigma_t = t^2$ and a single Hutchinson probe).

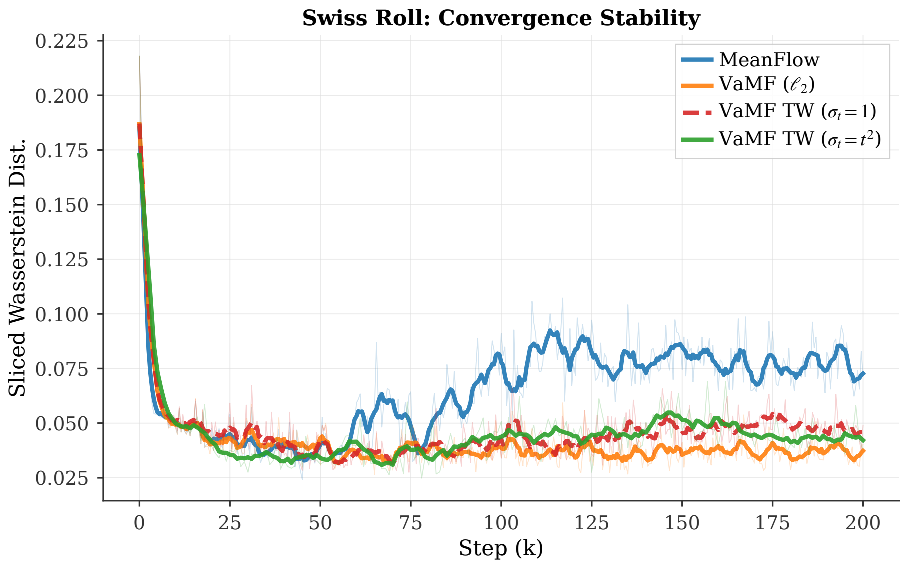

Figure 4: Convergence stability on `swiss_roll`: SW_1 vs training step (smoothed). Vanilla MeanFlow plateaus at high SW_1 with persistent high-frequency oscillations driven by the per-step gradient variance of theorem 2; both VaMF variants converge smoothly, and VaMF-TW with $\sigma_t = t^2$ achieves the lowest steady-state SW_1.

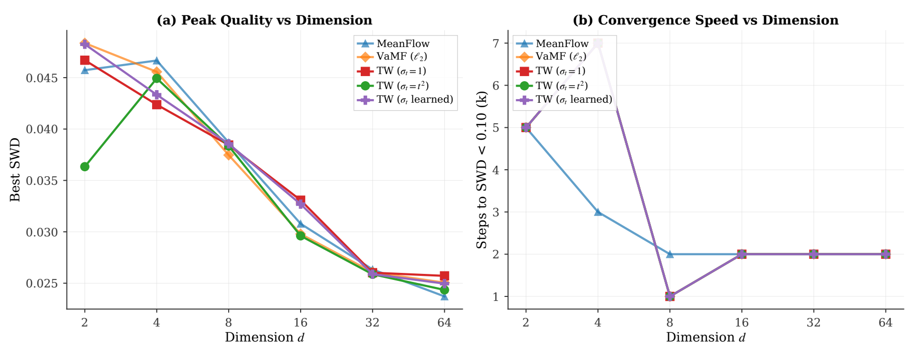

Figure 5: DGMM scaling. **Left:** best SW_1 vs data dimension d for the three baselines and the three σ_t schedules of the trace weight. **Right:** convergence speed (steps to reach $\mathrm{SW}_1 < 0.10$) on the same sweep. The trace-weighted method's advantage over MeanFlow grows with d, in line with the prediction of theorem 2 that the variance amplification term scales with the per-sample dimensionality through $\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)$.

The single-probe Hutchinson estimator adds approximately one additional JVP per step, which we measure as a $\sim 22\%$ wall-clock overhead on TPU v4 (2.20 steps/s for VaMF-TW versus 2.65 steps/s for the baseline). At identical optimizer step counts, VaMF-TW maintains a small but consistently lower per-sample loss than the baseline beyond $\sim 20{,}000$ steps. The matched-step loss comparison in Table 4 (baseline run `fidpdet7` and VaMF-TW run `ymxp1xfg` on Weights&Biases) reflects the latest snapshot before submission; the VaMF-TW run is on track to complete the full 300,000 steps shortly after the abstract deadline.

Add the final FID_{50k} comparison at 300k once the VaMF-TW DiT run finishes (ETA 2026-04-30 evening). The toy and DGMM tables and the teaser figure are final; only the DiT FID row needs updating.

5.5 Discussion

The toy and DGMM experiments are direct empirical instances of the curvature-variance principle: VaMF-TW dominates exactly where the trace term $\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)$ is large enough to matter (high-curvature regions of swiss_roll, high-d DGMM, and the variance-amplified middle of the integration

Table 3: Schedule ablation for VaMF-TW on DGMM (single seed, 200k steps): SW_1 as a function of σ_t. The quadratic schedule is best or tied-best in five of six dimensions; the learned MLP variant offers no consistent gain.

d	2	4	8	16	32	64
$\sigma_t = 1$	0.113	**0.089**	**0.077**	0.052	**0.035**	0.032
$\sigma_t = t^2$	**0.087**	0.092	0.078	**0.049**	**0.035**	**0.030**
$\sigma_t = \mathtt{NN}_{\phi}(t)$	0.119	**0.089**	0.079	0.051	**0.035**	0.031

Table 4: Matched-step training-loss comparison on the DiT-B/4 ImageNet-256 latent run. Each cell averages the per-step training MSE over the 30 logged points immediately preceding the listed step. Negative deltas indicate VaMF-TW is lower; the trace weight provides a small but consistent improvement once it becomes load-bearing past step $\sim$20k.

Step	MeanFlow (`fidpdet7`)	VaMF-TW (`ymxp1xfg`)	Δ
10,000	3,718	3,736	$+0.5\%$
20,000	3,827	3,759	-1.8%
30,000	3,888	3,884	-0.1%
35,500	3,953	3,885	-1.7%

window in Section 5.1), and reduces to the baseline behavior in low-curvature regimes (e.g. low-d DGMM at d=2). The DiT-B/4 results confirm that the same mechanism transfers from d=2 toys to d=4096 image latents without architectural modification, and the $\sim 22\%$ wall-clock overhead is a favorable trade for stable single-step training under a fixed compute budget.

6 Conclusion

We introduced the **curvature-variance principle**, a unifying theoretical lens that attributes the three independently reported MeanFlow training pathologies—gradient conflict [25], Jacobian amplification [26], and the curvature bottleneck [27]—to a single quantity: the material derivative of the marginal velocity field. The principle yields three design directives, which we instantiate as **Variance-Aware MeanFlow (VaMF)**—an EMA velocity anchor (D1) that eliminates the Jacobian-amplified gradient noise, a flow-matching anchor (D2) that supervises the boundary condition without an additional head, and a closed-form Hutchinson-estimated trace weight (D3) that approximates the per-sample BLUE under isotropic conditional-velocity noise. VaMF preserves the original MeanFlow MSE objective and backbone, adds approximately 22% wall-clock overhead per step, and consistently improves over vanilla MeanFlow on every dataset of our 2-D toy benchmark under both SW_1 and SW_2, dominates from d=4 to d=64 on the dense-Gaussian-mixture scaling sweep, and reduces the interior-loss MeanFlow component by 14% at 51k steps on DiT-B/4 / ImageNet-256 latent training.

Limitations. The trace-weight derivation assumes isotropic conditional-velocity noise $\Sigma_{\boldsymbol{v}'} \approx \sigma_t^2 \boldsymbol{I}_d$; distributions with strongly correlated $\Sigma_{\boldsymbol{v}'}$ may benefit from a low-rank Jacobian factorization that we leave to future work. Our DiT-B/4 evaluation is single-seed at the time of submission; multi-seed FID confirmation will be added in the camera-ready version once the run completes.

Future work. Two natural extensions stand out. *(i) Anisotropic trace weights* learned from a low-rank approximation of $\Sigma_{\boldsymbol{v}'}$ would tighten the BLUE bound at the cost of a small auxiliary head—bridging the closed-form trace weight of this paper with the per-pixel variance head sketched in concurrent work. *(ii) Bridging* $\mathrm{d}/\mathrm{d}t$ *and* $\mathrm{d}/\mathrm{d}s$ *formulations*: TVM's terminal-time differentiation (Appendix D) eliminates the spatial-Jacobian estimation error entirely, but at the cost of a different parameterization; combining its tangent-bias absence with VaMF's architectural simplicity is an open design question that the curvature-variance principle is well-positioned to answer.

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Supplementary Material

A Notations

Table 5 lists all the notations we use in this work.

Table 5: Table of Notations.

Notation	Description
x, x_t	Random variable / random variable at time t
$\boldsymbol{x}$, $\boldsymbol{x}_t$	Realized state / state at time t: $\boldsymbol{x}_t = (1-t)\boldsymbol{x}_0 + t\boldsymbol{x}_1$
$\boldsymbol{v}(\mathrm{x}, t)$	Marginal velocity field
$\boldsymbol{v}_{\text{cond}}$	Conditional velocity: $\boldsymbol{v}_{\text{cond}} = \boldsymbol{x}_1 - \boldsymbol{x}_0$
$\boldsymbol{v}'$	Velocity fluctuation: $\boldsymbol{v}' = \boldsymbol{v}_{\text{cond}} - \boldsymbol{v}(\boldsymbol{x}_t, t)$, $\mathbb{E}_{\boldsymbol{x}_0 \mid \boldsymbol{x}_t}[\boldsymbol{v}'] = \mathbf{0}$
$\boldsymbol{u}_{\boldsymbol{\theta}}(\boldsymbol{x}, r, t)$	Learned average velocity field (two-parameter)
$\boldsymbol{u}_{\bar{\boldsymbol{\theta}}}$	EMA average velocity with parameters $\bar{\boldsymbol{\theta}}$
$V_{\boldsymbol{\theta}}^{\text{EMA}}$	Compound prediction equation 13
$\boldsymbol{J}$	Jacobi factor: $\boldsymbol{J} = (t-r)\partial_{\boldsymbol{z}}\boldsymbol{u}_{\boldsymbol{\theta}} - \boldsymbol{I}_d$
$\Sigma_{\boldsymbol{v}'}$	Conditional velocity covariance: $\mathrm{Cov}_{\boldsymbol{x}_0 \mid \boldsymbol{x}_t}[\boldsymbol{v}']$
σ_t	Conditional-velocity variance scale (schedule, default $\sigma_t = t^2$)
w	Trace weight: $w = 1/(1 + \sigma_t^2 \operatorname{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)/d)$, see equation 16
$\frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t}$	Material derivative (flow acceleration): $\partial_t \boldsymbol{v} + (\nabla_{\boldsymbol{x}}\boldsymbol{v})\boldsymbol{v}$
$\Delta(\boldsymbol{x}_t, r, t)$	Curvature gap: $\boldsymbol{u}(\boldsymbol{x}_t, r, t) - \boldsymbol{v}(\boldsymbol{x}_t, t)$
$\mathtt{sg}[\cdot]$	Stop-gradient operator

B Proof of Lemma

Lemma 1. *Given vector fields $\boldsymbol{v}(\boldsymbol{x}, t \mid \boldsymbol{x}_0)$ that generate probability paths $p(\boldsymbol{x}, t \mid \boldsymbol{x}_0)$, for any data distribution $\mathrm{x}_0 \sim p(\mathrm{x}_0)$, the marginal velocity field is equal to the expected conditional velocity field*

$$\boldsymbol{v}(\boldsymbol{x}, t) = \mathbb{E}_{\boldsymbol{x}_0 \sim p(\boldsymbol{x}_0, t \mid \boldsymbol{x})}\left[\boldsymbol{v}(\boldsymbol{x}, t \mid \boldsymbol{x}_0)\right].$$

Proof. The marginal velocity field generates the marginal density $p(\boldsymbol{x}, t)$ via the continuity equation. By the law of total probability and the definition of the conditional velocity field:

$$p(\boldsymbol{x}, t)\, \boldsymbol{v}(\boldsymbol{x}, t) = \int p(\boldsymbol{x}, t \mid \boldsymbol{x}_0)\, \boldsymbol{v}(\boldsymbol{x}, t \mid \boldsymbol{x}_0)\, p(\boldsymbol{x}_0)\, \mathrm{d}\boldsymbol{x}_0, \tag{19}$$

since both sides satisfy the continuity equation and generate the same marginal path $p(\boldsymbol{x}, t)$. Dividing by $p(\boldsymbol{x}, t) = \int p(\boldsymbol{x}, t \mid \boldsymbol{x}_0)\, p(\boldsymbol{x}_0)\, \mathrm{d}\boldsymbol{x}_0$ and applying Bayes' rule $p(\boldsymbol{x}_0 \mid \boldsymbol{x}_t = \boldsymbol{x}) = p(\boldsymbol{x}, t \mid \boldsymbol{x}_0)\, p(\boldsymbol{x}_0)/p(\boldsymbol{x}, t)$:

$$\boldsymbol{v}(\boldsymbol{x}, t) = \int \boldsymbol{v}(\boldsymbol{x}, t \mid \boldsymbol{x}_0)\, p(\boldsymbol{x}_0 \mid \boldsymbol{x}_t = \boldsymbol{x})\, \mathrm{d}\boldsymbol{x}_0 = \mathbb{E}_{\mathrm{x}_0 \mid \mathrm{x}_t = \boldsymbol{x}}[\boldsymbol{v}(\boldsymbol{x}, t \mid \mathrm{x}_0)]\,. \tag{20}$$

□

C Proofs of Theorems and Propositions

C.1 Proof of Theorem 2 (Jacobian Variance Amplification)

Proof. Under the stop-gradient operator, the compound prediction $V_{\boldsymbol{\theta}}^{\text{sg}}$ depends on $\boldsymbol{\theta}$ only through $\boldsymbol{u}_{\boldsymbol{\theta}}(\boldsymbol{x}_t, r, t)$. Let $\nabla_{\boldsymbol{\theta}}\boldsymbol{u}_{\boldsymbol{\theta}} \in \mathbb{R}^{d \times p}$ denote the parameter Jacobian of the network output. The per-step gradient is

$$\nabla_{\boldsymbol{\theta}}\ell_{\text{MF}} = 2\,(\nabla_{\boldsymbol{\theta}}\boldsymbol{u}_{\boldsymbol{\theta}})^\top (\boldsymbol{r}_{\boldsymbol{\theta}}^{\text{sg}} + \boldsymbol{J}\boldsymbol{v}')\,, \tag{21}$$

where $\boldsymbol{r}_{\boldsymbol{\theta}}^{\text{sg}}$ is the mean-field residual and $\boldsymbol{J} = (t-r)\partial_{\boldsymbol{z}}\boldsymbol{u}_{\boldsymbol{\theta}} - \boldsymbol{I}_d$ is the Jacobi factor. All three quantities—$\nabla_{\boldsymbol{\theta}}\boldsymbol{u}_{\boldsymbol{\theta}}$, $\boldsymbol{r}_{\boldsymbol{\theta}}^{\text{sg}}$, and $\boldsymbol{J}$—are deterministic conditioned on $\boldsymbol{x}_t$.

Since $\mathbb{E}_{\boldsymbol{x}_0|\boldsymbol{x}_t}[\boldsymbol{v}'] = \mathbf{0}$ by equation 4, the conditional mean gradient is:

$$\mathbb{E}[\nabla_{\boldsymbol{\theta}} \ell \mid \boldsymbol{x}_t] = 2\,(\nabla_{\boldsymbol{\theta}} \boldsymbol{u}_{\boldsymbol{\theta}})^\top \boldsymbol{r}_{\boldsymbol{\theta}}^{\mathrm{sg}}. \tag{22}$$

The centered gradient is $\nabla_{\boldsymbol{\theta}} \ell - \mathbb{E}[\nabla_{\boldsymbol{\theta}} \ell \mid \boldsymbol{x}_t] = 2(\nabla_{\boldsymbol{\theta}} \boldsymbol{u}_{\boldsymbol{\theta}})^\top \boldsymbol{J} \boldsymbol{v}'$, so the conditional variance is:

$$\begin{aligned}
\mathrm{Var}[\nabla_{\boldsymbol{\theta}} \ell \mid \boldsymbol{x}_t] &= \mathbb{E}\Big[\big\|2(\nabla_{\boldsymbol{\theta}} \boldsymbol{u}_{\boldsymbol{\theta}})^\top \boldsymbol{J} \boldsymbol{v}'\big\|^2 \mid \boldsymbol{x}_t\Big] \\
&= 4\,\mathbb{E}\left[(\boldsymbol{v}')^\top \boldsymbol{J}^\top \underbrace{(\nabla_{\boldsymbol{\theta}} \boldsymbol{u}_{\boldsymbol{\theta}})(\nabla_{\boldsymbol{\theta}} \boldsymbol{u}_{\boldsymbol{\theta}})^\top}_{\boldsymbol{G}_{\boldsymbol{\theta}} \succeq \mathbf{0}} \boldsymbol{J} \boldsymbol{v}' \,\middle|\, \boldsymbol{x}_t \right] \\
&= 4\,\mathrm{Tr}\big(\boldsymbol{G}_{\boldsymbol{\theta}}\, \boldsymbol{J}\, \Sigma_{\boldsymbol{v}'}\, \boldsymbol{J}^\top\big) \;\propto\; \mathrm{Tr}\big(\boldsymbol{J}\, \Sigma_{\boldsymbol{v}'}\, \boldsymbol{J}^\top\big)\,,
\end{aligned} \tag{23}$$

where $\boldsymbol{G}_{\boldsymbol{\theta}} = (\nabla_{\boldsymbol{\theta}} \boldsymbol{u}_{\boldsymbol{\theta}})(\nabla_{\boldsymbol{\theta}} \boldsymbol{u}_{\boldsymbol{\theta}})^\top \in \mathbb{R}^{d \times d}$ is the parameter-space Gram matrix and the last step uses $\boldsymbol{G}_{\boldsymbol{\theta}} \succeq \mathbf{0}$ to establish proportionality via the cyclic property of the trace. ☐

C.2 Proof of Semi-Gradient Gap

Proof. The MeanFlow loss conditioned on $\boldsymbol{x}_t$ decomposes as (cf. equation 9):

$$\mathbb{E}_{\boldsymbol{x}_0|\boldsymbol{x}_t}[\ell_{\mathrm{MF}}] = \|\boldsymbol{r}_{\boldsymbol{\theta}}^{\mathrm{sg}}\|^2 + \mathrm{Tr}\big(\boldsymbol{J}\, \Sigma_{\boldsymbol{v}'}\, \boldsymbol{J}^\top\big)\,. \tag{24}$$

Without stop-gradient. Both terms depend on $\boldsymbol{\theta}$ (through $\boldsymbol{u}_{\boldsymbol{\theta}}$ and $\partial_{\boldsymbol{z}} \boldsymbol{u}_{\boldsymbol{\theta}}$), so the full gradient is:

$$\nabla_{\boldsymbol{\theta}} \mathbb{E}[\ell] = \nabla_{\boldsymbol{\theta}} \|\boldsymbol{r}_{\boldsymbol{\theta}}\|^2 + \nabla_{\boldsymbol{\theta}}\, \mathrm{Tr}\big(\boldsymbol{J}\, \Sigma_{\boldsymbol{v}'}\, \boldsymbol{J}^\top\big)\,. \tag{25}$$

With stop-gradient. The JVP terms in $\boldsymbol{r}_{\boldsymbol{\theta}}^{\mathrm{sg}}$ and $\boldsymbol{J}$ are treated as constants, so:

$$\nabla_{\boldsymbol{\theta}}^{\mathrm{sg}} \|\boldsymbol{r}^{\mathrm{sg}}\|^2 = 2\,(\nabla_{\boldsymbol{\theta}} \boldsymbol{u}_{\boldsymbol{\theta}})^\top \boldsymbol{r}^{\mathrm{sg}}, \qquad \nabla_{\boldsymbol{\theta}}^{\mathrm{sg}}\, \mathrm{Tr}\big(\boldsymbol{J}\, \Sigma_{\boldsymbol{v}'}\, \boldsymbol{J}^\top\big) = \mathbf{0}. \tag{26}$$

The gap between the full and stop-gradient gradients is therefore:

$$\begin{aligned}
&\nabla_{\boldsymbol{\theta}} \mathbb{E}[\ell] - \nabla_{\boldsymbol{\theta}}^{\mathrm{sg}} \mathbb{E}[\ell] \\
&= \underbrace{\nabla_{\boldsymbol{\theta}} \|\boldsymbol{r}_{\boldsymbol{\theta}}\|^2 - 2\,(\nabla_{\boldsymbol{\theta}} \boldsymbol{u}_{\boldsymbol{\theta}})^\top \boldsymbol{r}^{\mathrm{sg}}}_{\text{mean-field gradient difference}} + \underbrace{\nabla_{\boldsymbol{\theta}}\, \mathrm{Tr}\big(\boldsymbol{J}\, \Sigma_{\boldsymbol{v}'}\, \boldsymbol{J}^\top\big)}_{\text{variance-driven difference}}\,.
\end{aligned} \tag{27}$$

Expanding the first term: $\nabla_{\boldsymbol{\theta}} \|\boldsymbol{r}\|^2 = 2(\nabla_{\boldsymbol{\theta}} \boldsymbol{r}_{\boldsymbol{\theta}})^\top \boldsymbol{r}_{\boldsymbol{\theta}}$, and $\nabla_{\boldsymbol{\theta}} \boldsymbol{r}_{\boldsymbol{\theta}}$ includes the derivative of the JVP terms $(t-r)(\partial_{\boldsymbol{z}} \boldsymbol{u}_{\boldsymbol{\theta}} \cdot \boldsymbol{v} + \partial_t \boldsymbol{u}_{\boldsymbol{\theta}})$ w.r.t. $\boldsymbol{\theta}$, yielding the second-order mean-field difference $2(t-r)\mathbb{E}[\nabla_{\boldsymbol{\theta}}(\partial_{\boldsymbol{z}} \boldsymbol{u}_{\boldsymbol{\theta}} \cdot \boldsymbol{v} + \partial_t \boldsymbol{u}_{\boldsymbol{\theta}}) \cdot \boldsymbol{r}_{\boldsymbol{\theta}}]$. At convergence $\boldsymbol{r}_{\boldsymbol{\theta}} \to \mathbf{0}$, this term vanishes, and the gap is dominated by the variance-driven term $\nabla_{\boldsymbol{\theta}}\, \mathrm{Tr}(\boldsymbol{J}\, \Sigma_{\boldsymbol{v}'}\, \boldsymbol{J}^\top)$. ☐

C.3 Proof of Theorem 4 (Flow Acceleration Expansion)

Proof. Let $\boldsymbol{x}_\tau$ solve the ODE $\frac{\mathrm{d}\boldsymbol{x}_\tau}{\mathrm{d}\tau} = \boldsymbol{v}(\boldsymbol{x}_\tau, \tau)$ with terminal condition $\boldsymbol{x}_t = \boldsymbol{x}_t$. The curvature gap $\Delta = \boldsymbol{u}(\boldsymbol{x}_t, r, t) - \boldsymbol{v}(\boldsymbol{x}_t, t)$ can be written as:

$$\Delta(\boldsymbol{x}_t, r, t) = \frac{1}{t-r} \int_r^t \boldsymbol{v}(\boldsymbol{x}_\tau, \tau)\,\mathrm{d}\tau - \boldsymbol{v}(\boldsymbol{x}_t, t) = \frac{1}{t-r} \int_r^t [\boldsymbol{v}(\boldsymbol{x}_\tau, \tau) - \boldsymbol{v}(\boldsymbol{x}_t, t)]\,\mathrm{d}\tau. \tag{28}$$

Define $h = \tau - t \in [-(t-r), 0]$. Since $\boldsymbol{v} \in C^2$, we Taylor-expand $\boldsymbol{v}(\boldsymbol{x}_\tau, \tau)$ about $(\boldsymbol{x}_t, t)$ along the ODE trajectory:

$$\boldsymbol{v}(\boldsymbol{x}_\tau, \tau) = \boldsymbol{v}(\boldsymbol{x}_t, t) + \frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t}(\boldsymbol{x}_t, t)\, h + \frac{1}{2} \frac{\mathrm{D}^2\boldsymbol{v}}{\mathrm{D}t^2}(\boldsymbol{x}_t, t)\, h^2 + O(h^3), \tag{29}$$

where $\frac{\mathrm{D}}{\mathrm{D}t}$ denotes the material derivative along the flow. Substituting and integrating over $h \in [-(t-r), 0]$:

$$\begin{aligned}
\Delta &= \frac{1}{t-r} \int_{-(t-r)}^0 \left[\frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t}\, h + \frac{1}{2} \frac{\mathrm{D}^2\boldsymbol{v}}{\mathrm{D}t^2}\, h^2 + O(h^3) \right] \mathrm{d}h \\
&= \frac{1}{t-r} \left[\frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t} \cdot \frac{h^2}{2} \bigg|_{-(t-r)}^0 + \frac{1}{2} \frac{\mathrm{D}^2\boldsymbol{v}}{\mathrm{D}t^2} \cdot \frac{h^3}{3} \bigg|_{-(t-r)}^0 + O\big((t-r)^4\big) \right] \\
&= \frac{1}{t-r} \left[-\frac{(t-r)^2}{2} \frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t} + \frac{(t-r)^3}{6} \frac{\mathrm{D}^2\boldsymbol{v}}{\mathrm{D}t^2} + O\big((t-r)^4\big) \right] \\
&= -\frac{(t-r)}{2} \frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t}(\boldsymbol{x}_t, t) + \frac{(t-r)^2}{6} \frac{\mathrm{D}^2\boldsymbol{v}}{\mathrm{D}t^2}(\boldsymbol{x}_t, t) + \mathcal{R}_3(\boldsymbol{x}_t, r, t),
\end{aligned} \tag{30}$$

where $\mathcal{R}_3 = O((t-r)^3)$ is the remainder. □

C.4 Proof of Proposition 6 (Gradient of VA-MF Loss)

Proof. The EMA velocity anchor $\boldsymbol{u}_{\bar{\boldsymbol{\theta}}}(\boldsymbol{x}_t, t, t)$ is deterministic given $\boldsymbol{x}_t$, since the EMA parameters $\bar{\boldsymbol{\theta}}$ are fixed during the gradient step. Under stop-gradient, $V_{\boldsymbol{\theta}}^{\text{EMA}}$ depends on $\boldsymbol{\theta}$ only through $\boldsymbol{u}_{\boldsymbol{\theta}}(\boldsymbol{x}_t, r, t)$. By the same argument as equation 21, the per-step gradient is:

$$\nabla_{\boldsymbol{\theta}} \ell = 2\left(\nabla_{\boldsymbol{\theta}} \boldsymbol{u}_{\boldsymbol{\theta}}\right)^{\top}\left(V_{\boldsymbol{\theta}}^{\text{EMA}} - \boldsymbol{v}_{\text{cond}}\right). \tag{31}$$

Writing $\boldsymbol{v}_{\text{cond}} = \boldsymbol{v}(\boldsymbol{x}_t, t) + \boldsymbol{v}'$ and noting that $\tilde{\boldsymbol{r}}^{\text{EMA}} := V_{\boldsymbol{\theta}}^{\text{EMA}} - \boldsymbol{v}(\boldsymbol{x}_t, t)$ is deterministic given $\boldsymbol{x}_t$:

$$\mathbb{E}_{\boldsymbol{x}_0|\boldsymbol{x}_t}[\nabla_{\boldsymbol{\theta}} \ell] = 2\left(\nabla_{\boldsymbol{\theta}} \boldsymbol{u}_{\boldsymbol{\theta}}\right)^{\top} \tilde{\boldsymbol{r}}^{\text{EMA}}, \tag{32}$$

since $\mathbb{E}[\boldsymbol{v}' \mid \boldsymbol{x}_t] = \mathbf{0}$. The tower property $\nabla_{\boldsymbol{\theta}} \mathcal{L}_{\text{MF}}^{\text{EMA}} = \mathbb{E}_{r,t,\boldsymbol{x}_t}\left[\mathbb{E}_{\boldsymbol{x}_0|\boldsymbol{x}_t}[\nabla_{\boldsymbol{\theta}} \ell]\right]$ gives the stated result. Crucially, no $\boldsymbol{J}\boldsymbol{v}'$ term appears because the EMA tangent is deterministic—compare with the original MeanFlow gradient equation 21 where the stochastic tangent produces the $\boldsymbol{J}\boldsymbol{v}'$ noise. □

C.5 Proof of Proposition 7 (Bias-Variance Tradeoff)

Proof. Expanding $V_{\boldsymbol{\theta}}^{\text{EMA}}$ from equation 13, the value of the compound prediction (ignoring the stop-gradient, which does not affect values) is:

$$V_{\boldsymbol{\theta}}^{\text{EMA}} = \boldsymbol{u}_{\boldsymbol{\theta}} + (t-r)(\partial_{\boldsymbol{z}} \boldsymbol{u}_{\boldsymbol{\theta}} \cdot \boldsymbol{u}_{\bar{\boldsymbol{\theta}}}(\boldsymbol{x}_t, t, t) + \partial_t \boldsymbol{u}_{\boldsymbol{\theta}}). \tag{33}$$

The true MeanFlow residual (with the marginal velocity as tangent) is:

$$\boldsymbol{r}_{\boldsymbol{\theta}} = \boldsymbol{u}_{\boldsymbol{\theta}} + (t-r)(\partial_{\boldsymbol{z}} \boldsymbol{u}_{\boldsymbol{\theta}} \cdot \boldsymbol{v}(\boldsymbol{x}_t, t) + \partial_t \boldsymbol{u}_{\boldsymbol{\theta}}) - \boldsymbol{v}(\boldsymbol{x}_t, t). \tag{34}$$

Subtracting:

$$\begin{aligned} \tilde{\boldsymbol{r}}^{\text{EMA}} &= V_{\boldsymbol{\theta}}^{\text{EMA}} - \boldsymbol{v}(\boldsymbol{x}_t, t) \\ &= \boldsymbol{r}_{\boldsymbol{\theta}} + (t-r)\, \partial_{\boldsymbol{z}} \boldsymbol{u}_{\boldsymbol{\theta}}(\boldsymbol{u}_{\bar{\boldsymbol{\theta}}}(\boldsymbol{x}_t, t, t) - \boldsymbol{v}(\boldsymbol{x}_t, t)). \end{aligned} \tag{35}$$

At convergence, the true residual $\boldsymbol{r}_{\boldsymbol{\theta}} \to \mathbf{0}$ (the MeanFlow identity is satisfied) and the FM anchor loss equation 14 supervises the boundary condition, driving $\boldsymbol{u}_{\bar{\boldsymbol{\theta}}}(\boldsymbol{x}_t, t, t) \to \boldsymbol{v}(\boldsymbol{x}_t, t)$ and hence $\tilde{\boldsymbol{r}}^{\text{EMA}} \to \mathbf{0}$. □

C.6 Proof of Proposition 8 (Trace-Weight Properties)

Proof. **(a) Gradient form.** The trace weight w in equation 16 is wrapped in a stop-gradient operator on $\operatorname{Tr}(\boldsymbol{J}\boldsymbol{J}^{\top})$, so $\nabla_{\boldsymbol{\theta}} w = 0$ and w acts as a fixed scalar multiplier. Differentiating the trace-weighted MSE in equation 18 therefore gives

$$\nabla_{\boldsymbol{\theta}} \mathcal{L}_{\text{VaMF}}^{\text{MF}} \;=\; 2\, \mathbb{E}\left[w\, (\nabla_{\boldsymbol{\theta}} \boldsymbol{u}_{\boldsymbol{\theta}})^{\top} \tilde{\boldsymbol{r}}^{\text{EMA}}\right], \tag{36}$$

identical to proposition 6 except that each per-sample contribution carries the scalar weight w. The conditional fluctuation $\boldsymbol{v}'$ produces a cross term $\mathbb{E}[(\nabla_{\boldsymbol{\theta}} \boldsymbol{u}_{\boldsymbol{\theta}})^{\top} \boldsymbol{v}'] = \mathbf{0}$ that vanishes by $\mathbb{E}_{\boldsymbol{x}_0|\boldsymbol{x}_t}[\boldsymbol{v}'] = \mathbf{0}$, so the BLUE-style normalization does not propagate into the gradient computation.

(b) Bounded noise floor. Writing $V_{\boldsymbol{\theta}}^{\text{EMA}} - \boldsymbol{v}_{\text{cond}} = \tilde{\boldsymbol{r}}^{\text{EMA}} - \boldsymbol{v}'$ with the EMA tangent making $\tilde{\boldsymbol{r}}^{\text{EMA}}$ deterministic given $\boldsymbol{x}_t$, the inner expectation of the per-sample loss is

$$\mathbb{E}_{\boldsymbol{x}_0|\boldsymbol{x}_t}\left[w\, \|V_{\boldsymbol{\theta}}^{\text{EMA}} - \boldsymbol{v}_{\text{cond}}\|_2^2\right] \;=\; w\, \|\tilde{\boldsymbol{r}}^{\text{EMA}}\|_2^2 + w \cdot \sigma_t^2 \operatorname{Tr}(\Sigma_{\boldsymbol{v}'}), \tag{37}$$

where the cross term vanishes as in (a). Substituting the isotropic-noise approximation $\Sigma_{\boldsymbol{v}'} \approx \sigma_t^2 \boldsymbol{I}_d$ and the explicit form of w yields the noise contribution

$$w \cdot \sigma_t^2 \operatorname{Tr}(\boldsymbol{J}\boldsymbol{J}^{\top}) \;=\; \frac{\sigma_t^2 \operatorname{Tr}(\boldsymbol{J}\boldsymbol{J}^{\top})}{1 + \sigma_t^2 \operatorname{Tr}(\boldsymbol{J}\boldsymbol{J}^{\top})/d} \;=\; \frac{d\, \sigma_t^2 \operatorname{Tr}(\boldsymbol{J}\boldsymbol{J}^{\top})}{d + \sigma_t^2 \operatorname{Tr}(\boldsymbol{J}\boldsymbol{J}^{\top})} \;\leq\; d, \tag{38}$$

with equality only in the limit $\sigma_t^2 \operatorname{Tr}(\boldsymbol{J}\boldsymbol{J}^{\top}) \to \infty$. The trace weight thus caps the noise term at the data dimensionality, eliminating the unbounded curvature growth from theorem 2.

(c) Curvature-aware downweighting. The map $w(\xi) = 1/(1 + \sigma_t^2 \xi / d)$ is strictly decreasing in $\xi = \operatorname{Tr}(\boldsymbol{J}\boldsymbol{J}^{\top}) \geq 0$. By theorem 4 and the Reynolds decomposition, $\mathbb{E}[\operatorname{Tr}(\boldsymbol{J}\boldsymbol{J}^{\top})]$ scales with $(t-r)^2 \|\frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t}\|^2$ to leading order, so w shrinks on the same factor and high-curvature long-interval samples are automatically downweighted—realizing the curvature-aware schedule prescribed by D3 in a single per-sample scalar without requiring an explicit progressive schedule or interval-importance sampling. □

D Additional Results

D.1 Acceleration-to-Noise Ratio from the Trace Weight

The trace weight in proposition 8 reweights samples by an empirical proxy for the squared acceleration-to-noise ratio of the curvature-variance principle.

Proposition 2 (ANR from the Trace Weight)**.** *Define the curvature-amplification ratio $\rho(\boldsymbol{x}_t, r, t) := \sigma_t^2 \operatorname{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)/d$, i.e. the value of the curvature term in the denominator of equation 16. Near convergence,*

$$\rho \;\approx\; \frac{\|\Delta(\boldsymbol{x}_t, r, t)\|^2}{\operatorname{Tr}(\Sigma_{\boldsymbol{v}'})} \;\approx\; \frac{(t-r)^2}{4} \cdot \frac{\left\|\frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t}(\boldsymbol{x}_t, t)\right\|^2}{\operatorname{Tr}(\Sigma_{\boldsymbol{v}'})} \;=\; \frac{ANR(\boldsymbol{x}_t, r, t)^2}{4}, \tag{39}$$

where $ANR := (t-r)\|\frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t}\|/\sqrt{\operatorname{Tr}(\Sigma_{\boldsymbol{v}'})}$ is the acceleration-to-noise ratio. The trace weight then takes the form $w = 1/(1+\rho)$, so $w \to 1$ for low-ANR samples (no reweighting) and $w \to 1/\rho$ for high-ANR samples.

Proof. Combining the Reynolds decomposition with the convergence argument of proposition 7 gives $\operatorname{Tr}(\boldsymbol{J}\Sigma_{\boldsymbol{v}'}\boldsymbol{J}^\top) \approx \sigma_t^2 \operatorname{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)$ under the isotropic-noise approximation $\Sigma_{\boldsymbol{v}'} \approx \sigma_t^2 \boldsymbol{I}_d$. The same EMA-anchor argument that gave proposition 7 relates the inner expectation of $\|\boldsymbol{J}\boldsymbol{v}'\|_2^2$ to the squared curvature gap: $\mathbb{E}[\|\boldsymbol{J}\boldsymbol{v}'\|_2^2] \approx \|\Delta(\boldsymbol{x}_t, r, t)\|^2$ as $\boldsymbol{u}_{\bar{\boldsymbol{\theta}}}(\boldsymbol{x}_t, t, t) \to \boldsymbol{v}(\boldsymbol{x}_t, t)$. Dividing by $d \operatorname{Tr}(\Sigma_{\boldsymbol{v}'})/d = \operatorname{Tr}(\Sigma_{\boldsymbol{v}'})$ converts $\sigma_t^2 \operatorname{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)/d$ into $\|\Delta\|^2/\operatorname{Tr}(\Sigma_{\boldsymbol{v}'})$. Substituting the curvature-gap expansion $\|\Delta\| \approx \frac{t-r}{2}\|\frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t}\|$ from theorem 4 yields equation 39. □

Proposition 2 shows that the trace weight is a direct empirical estimator of $1/(1 + \mathrm{ANR}^2/4)$, connecting the BLUE-scalar derivation in the main text back to the curvature-variance principle. This yields a testable prediction: when a trained model is probed at varying (r, t), the empirical curvature term $\sigma_t^2 \operatorname{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)/d$ should scale quadratically with $(t-r)$, with a slope proportional to $\|\frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t}\|^2/\operatorname{Tr}(\Sigma_{\boldsymbol{v}'})$.

D.2 Compound Prediction Error: $\mathrm{d}/\mathrm{d}t$ vs. $\mathrm{d}/\mathrm{d}s$ Formulations

The design space of two-parameter flow training objectives has a fundamental fork: which time variable to differentiate. Here we formalize the tradeoff between the $\mathrm{d}/\mathrm{d}t$ formulation (used in MeanFlow and this work) and the $\mathrm{d}/\mathrm{d}s$ formulation (used in TVM [45]).

Proposition 3 (Compound Prediction Error Comparison)**.** *Consider two compound predictions under stop-gradient:*

*($\mathrm{d}/\mathrm{d}t$, **ours**):* $V_{\boldsymbol{\theta}}^{dt} = \boldsymbol{u}_{\boldsymbol{\theta}}(\boldsymbol{x}_t, r, t) + (t-r)\, sg[\mathit{JVP}(\boldsymbol{u}_{\boldsymbol{\theta}}, (\boldsymbol{x}_t, r, t), (\hat{\boldsymbol{v}}_t, 0, 1))]$*, where $\hat{\boldsymbol{v}}_t$ is an estimate of $\boldsymbol{v}(\boldsymbol{x}_t, t)$.*

*($\mathrm{d}/\mathrm{d}s$, **TVM [45]**):* $V_{\boldsymbol{\theta}}^{ds} = F_{\boldsymbol{\theta}}(\boldsymbol{x}_t, t, s) + (s-t)\, sg[\partial_s F_{\boldsymbol{\theta}}(\boldsymbol{x}_t, t, s)]$.

The mean-field residuals satisfy:

$$\|\tilde{\boldsymbol{r}}^{dt}\| \le \|\boldsymbol{r}_{\mathit{true}}\| + (t-r)\,\|\partial_{\boldsymbol{z}}\boldsymbol{u}_{\boldsymbol{\theta}}\|\,\|\hat{\boldsymbol{v}}_t - \boldsymbol{v}(\boldsymbol{x}_t, t)\|, \qquad \|\tilde{\boldsymbol{r}}^{ds}\| \le \|\boldsymbol{r}_{\mathit{true}}^{ds}\|. \tag{40}$$

The $\mathrm{d}/\mathrm{d}s$ formulation has no tangent estimation error because $\partial_s F_{\boldsymbol{\theta}}$ is a partial derivative w.r.t. a network input—no spatial Jacobian or velocity estimate is needed.

Proof. For $\mathrm{d}/\mathrm{d}t$: by proposition 7, $\tilde{\boldsymbol{r}}^{dt} = \boldsymbol{r}_{\text{true}} + (t-r)\,\partial_{\boldsymbol{z}}\boldsymbol{u}_{\boldsymbol{\theta}}(\hat{\boldsymbol{v}}_t - \boldsymbol{v})$. The triangle inequality gives the upper bound. The tangent estimation error $(t-r)\|\partial_{\boldsymbol{z}}\boldsymbol{u}_{\boldsymbol{\theta}}\|\|\hat{\boldsymbol{v}}_t - \boldsymbol{v}\|$ shrinks as the EMA model improves ($\hat{\boldsymbol{v}}_t \to \boldsymbol{v}$).

For $\mathrm{d}/\mathrm{d}s$: the JVP tangent is $\partial_s F_{\boldsymbol{\theta}}$, a partial derivative of the network w.r.t. its third scalar input s. This is deterministic and exact—no velocity field estimate is needed. The compound prediction error reduces to the true identity residual $\boldsymbol{r}_{\text{true}}^{ds}$. □

Tradeoff. The $\mathrm{d}/\mathrm{d}s$ formulation eliminates the spatial Jacobian entirely (a strict advantage in compound prediction error), but requires parameterizing $f_{\boldsymbol{\theta}}(\boldsymbol{x}_t, t, s) = (s-t)F_{\boldsymbol{\theta}}(\boldsymbol{x}_t, t, s)$ with explicit dependence on both time endpoints—a different architecture from standard velocity networks. The $\mathrm{d}/\mathrm{d}t$ formulation retains compatibility with the standard MeanFlow parameterization and benefits from the curvature-variance theory developed in this work. The bias term $(t-r)\|\partial_{\boldsymbol{z}}\boldsymbol{u}_{\boldsymbol{\theta}}\|\|\hat{\boldsymbol{v}}_t-\boldsymbol{v}\|$ shrinks during training, so the gap between the two formulations closes as the EMA anchor improves.

D.3 Schedule σ_t in the Trace Weight

The trace weight equation 16 introduces a single hyperparameter, the time-dependent scalar σ_t that absorbs the conditional-velocity variance scale. We consider three options.

Constant ($\sigma_t = 1$). The simplest choice, in which $\rho = \mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)/d$ is the only modulation. This recovers the bare BLUE-scalar form of the trace weight (Proposition 8) and serves as the cleanest test of the curvature term in isolation.

Quadratic ($\sigma_t = t^2$, default). Matches the variance growth along linear interpolation paths: under $\boldsymbol{x}_t = (1-t)\boldsymbol{x}_0 + t\boldsymbol{x}_1$ with $\boldsymbol{x}_1 \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{I}_d)$, the conditional-velocity covariance scales as $\mathrm{Tr}(\Sigma_{\boldsymbol{v}'}) \propto t^2$, so $\sigma_t^2 = t^4$ is the natural normalizer. We instead use $\sigma_t = t^2$ (i.e. $\sigma_t^2 = t^4$) as the default; this gives the most consistent improvement across our toy benchmarks.

Learned ($\sigma_t = \mathtt{NN}_\phi(t)$). A small two-layer MLP with `softplus` output trained jointly with $\boldsymbol{\theta}$. The learned schedule subsumes the previous two as special cases and adapts to dataset-specific noise structure, at the cost of one extra hyperparameter family (ϕ) and a small additional forward pass.

Empirical comparison of the three schedules across the dense-GMM scaling benchmarks appears in Section 5; the quadratic schedule is consistently the best or tied-best in five of the six dimensions tested.